The Paired-Delta Protocol: A Within-Call Reference Architecture for Observer QRNG Research

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Abstract

Experiments testing subtle effects on quantum random number generators are difficult to interpret when any condition-related signal is small relative to ordinary system variation. The Paired-Delta Protocol (PDP) was developed to increase the information available from each acquisition by retaining both a Subject stream and a same-call Paired Control Stream (PCS). This supports conventional Direct analysis, within-call Paired-differential analysis, and Relational analysis of joint-stream behavior. Study 1 piloted the architecture across Human, AI, and Baseline conditions. No clear paired mean hit-rate difference was detected, but an exploratory ordering-sensitive pattern showed how Subject, PCS, and Baseline could contribute differently to the same observed contrast. Exploratory relational analyses also identified candidate patterns in local correlation variability and temporal alignment.

Preregistered artifact-injection and held-out synthetic pattern-recovery tests showed that the implemented pipeline could recover the statistical structures it was designed to detect. A separate validation study using a real, controlled perturbation and switching between independent QRNG providers rather than a simulated disturbance, found no detectable provider-driven difference, though not yet at a resolution sufficient to certify equivalence. PDP therefore provides greater diagnostic resolution for investigating subtle condition-related differences.

1. Introduction

1.1 The Credibility Problem in Observer–QRNG Research

Decades of experiments testing whether human intention influences quantum or electronic random number generators have produced a large but contested body of data (e.g., Jahn et al., 1997). The central dispute concerns the credibility and reproducibility of very small reported effects. Bösch, Steinkamp, and Boller's (2006) meta-analysis of 380 studies found a statistically significant but extremely small and highly heterogeneous overall effect whose magnitude decreased as sample size increased. The ensuing exchange (Radin, Nelson, Dobyns, & Houtkooper, 2006; Wilson & Shadish, 2006) disputed the interpretation but not the underlying concern that small effects in this paradigm are difficult to separate from the combined influence of sampling variation, analytic flexibility, selective reporting, and other sources of systematic error.

More recent pre-registered work reported Bayesian evidence against a mean-level micro-psychokinetic effect (Maier, Dechamps, & Pflitsch, 2018). An exploratory analysis in the same study found an oscillating pattern across data collection. This does not establish an observer effect, but it shows that a null mean does not rule out other kinds of structure in the data. The harder question is how to tell whether an unusual pattern reflects the experimental condition or ordinary variation in the measurement system. That problem becomes especially difficult when only one stream from each acquisition is retained. PDP was developed to add information at that level by keeping a same-call companion stream alongside the Subject stream.

Statistical power alone does not resolve this problem. Steiglitz (2023) has argued that as sample size increases, ordinary sampling error decreases while uncertainty arising from imperfect RNG calibration may persist. Under those circumstances, sufficiently precise experiments can become sensitive to small systematic deviations in the apparatus as well as to any genuine condition-related effect. Calibration, adequate power, and experimental controls therefore address different sources of uncertainty and cannot substitute for one another.

The distinction among reference measurements is particularly important. Calibration runs characterize the behavior of the physical random system. An empirical Baseline condition provides a between-condition reference collected without the observer-related experimental condition. Conventional control runs can therefore reveal important apparatus or session-level behavior, but when control observations are acquired at a different time they cannot provide a matched reference for the particular acquisition being evaluated. PDP adds a third form of reference. The Paired Control Stream (PCS) is generated from the same QRNG call as the Subject stream. Because Subject and PCS share the immediate acquisition history, they can be compared within each call. The resulting paired difference can reduce contributions expressed similarly in both selected stream statistics, while separate examination of both components shows whether an observed contrast is confined primarily to Subject, appears in PCS as well, or is formed by movement in both.

Contributions that affect Subject and PCS unequally, vary across the acquisition, interact differently with the selected statistic, or arise from other unmodeled processes may survive subtraction. Conversely, a genuine condition-related change affecting both stream statistics similarly could also be attenuated or removed by the paired difference. The PCS therefore provides conditional common-mode control and diagnostic information rather than a universally preferable outcome measure.

These considerations identify two related but distinct problems.

1.1.1 Problem 1: Cross-study credibility

The first problem concerns the research process across studies. Undisclosed analytic choices, outcome-dependent stopping, and selective publication can produce small apparent effects across a literature. The accepted remedies are procedural. They include adequate calibration, pre-registration of hypotheses and analyses, locked datasets, and appropriate treatment of multiplicity and dependence. Analytic decisions should be fixed before outcomes are known, and results should be reported transparently regardless of direction.

The pilot reported here addresses this problem only partially. Metrics were implemented in the experimental code before data collection, and the completed dataset was cryptographically frozen before analysis, but Study 1 itself was exploratory and was not preregistered. Its empirical patterns are therefore treated as candidate findings rather than confirmatory evidence. Prospective evaluation of the observer-related candidate requires new outcome data collected under a preregistered analysis plan.

1.1.2 Problem 2: Within-study artifact ambiguity

The second problem exists even in a well-powered and well-reported individual study. Expected deviations may be small relative to variation associated with hardware behavior, calibration, provider characteristics, timing, environmental conditions, or other features of data acquisition. Increasing sample size raises sensitivity to systematic contributions as well as to genuine signals.

A single retained stream also cannot show whether a deviation is shared across or differs between parts of the same acquisition. It cannot reveal whether it is confined to that stream, shared with another portion of the same call, or accompanied by a change in the relationship between the two.

In distributed RNG networks such as the Global Consciousness Project, the project's own researchers have identified electromagnetic interference and other environmental factors as plausible alternative explanations for observed deviations (Nelson & Bancel, 2011). Conventional control data can help identify this kind of variation, but observations collected at a different time cannot provide a matched reference for the acquisition in which the deviation occurred.

PDP addresses this ambiguity by retaining a same-call reference. A prespecified within-call comparison can reduce contributions expressed equally in both selected stream statistics while retaining contributions that affect them differently. It does not identify the cause of a surviving difference, and contributions that affect the two streams unequally can remain.

A genuine condition-related change that affects both streams similarly could also be reduced by the subtraction.

The pilot tested this operating principle using controlled artifact injections. Contributions applied equally to both stream statistics produced little systematic displacement of the paired difference, while stream-specific and unequal contributions remained detectable. These tests showed that the pipeline behaved as intended under the constructed disturbances examined. They did not have sufficient resolution to establish cancellation at the scale of effects reported in the observer-RNG literature, and they did not establish removal of naturally occurring or unknown artifacts.

Classification rules developed from the exploratory pilot were subsequently tested in a separate preregistered held-out synthetic pattern-recovery study. Because the synthetic conditions were deliberately matched to the tested statistics and were stronger than the empirical relational candidates, that exercise established implementation and recovery within the specified constructions rather than diagnostic validity. It therefore tested whether the pipeline could recover known patterns under controlled conditions, not whether it had sufficient resolution to detect effects on the scale expected in the observer-RNG literature. Study 2 will provide prospective outcome data for prespecified observer-related estimands (Section 4.4).

A single retained stream can be analyzed with any number of metrics and transformations. Each provides another description of the same underlying data, but none adds information about how that stream relates to another part of the same acquisition. No transformation applied to one stream can show whether a pattern is confined to that stream, shared with a companion portion of the same draw, or organized differently across the two. PDP adds information that is different in kind. It retains conventional Direct analysis, adds a Paired-differential comparison between Subject and PCS, and makes properties of the relationship between the two streams directly measurable through Relational analysis. The Paired-differential and Relational quantities do not exist in a single-stream design. These capabilities are introduced in Section 1.5 and demonstrated in Sections 3 and 4.

1.2 The Paired-Delta Architecture

The Paired-Delta Protocol is a within-call reference architecture for experiments in which small differences may be difficult to distinguish from ordinary variation in the measurement system. Each acquisition yields a Subject stream and Paired Control Stream (PCS) from the same QRNG call. PDP does not require a particular stream length, partition rule, QRNG provider, or metric. These choices are specified in advance and calibrated for the intended analysis.

The same-call data support three forms of analysis. Direct analysis examines either stream separately and preserves the conventional single-stream comparison. Paired-differential analysis compares Subject and PCS within the same acquisition. Relational analysis examines properties of the pair itself, including how the two streams vary together. These analyses use the same underlying calls and are not statistically independent, but they do answer different questions.

PDP distinguishes two collected reference types. The PCS is the within-call reference generated alongside the Subject stream. Baseline is the empirical between-condition reference collected without the observer-related experimental condition. A theoretical chance value may provide an additional reference for some metrics, although the appropriate reference point can depend on the metric and its calibration.

For a selected stream-level measure M, the paired difference is

$$\Delta_M = M_{Subject} - M_{PCS}.$$

The paired comparison can reduce a contribution when it changes both selected stream statistics equally. Contributions that affect the streams unequally, vary across the acquisition, or interact differently with the selected metric may remain. The same limitation applies to a genuine condition-related change that affects both stream statistics similarly. Paired-differential analysis therefore provides conditional common-mode control rather than general protection from artifacts.

Conceptually, this resembles differential measurement in physical instrumentation, where two channels sharing a common source or measurement environment are compared so that contributions expressed similarly in both can be reduced while differential contributions remain. Interferometric systems provide one familiar example of this broader logic. The analogy is architectural rather than physical. Subject and PCS are partitions of recorded QRNG output and do not constitute interfering quantum pathways.

This is also why Paired does not replace Direct analysis. A condition-related change could appear in both Subject and PCS and be attenuated or cancelled by subtraction. Direct analysis retains sensitivity to such a whole-call or shared change, while the paired and separate PCS comparisons help show how the observed contrast is composed.

Paired-differential analysis also carries a precision cost. When the two stream statistics are independent and have similar variability, their difference is more variable than either stream statistic alone. PDP therefore retains the direct stream values rather than treating the paired difference as a universally preferable endpoint.

Relational analysis is different in kind from both Direct and Paired-differential analysis. Direct analysis concerns a property of one stream. Paired-differential analysis concerns the difference between two stream-level properties. Relational analysis concerns a property defined by the pair itself. For example, two streams could show similar marginal means across conditions while differing in their association, temporal alignment, or stability. These quantities require two defined streams and are measured directly in PDP from the paired streams generated within the same acquisition.

1.3 Study 1 as a Pilot Implementation

Study 1 was a pilot implementation of the Paired-Delta architecture. Its settings were chosen to make the study feasible with limited access to paid participants, API calls, and longer acquisitions. The pilot used 301-bit QRNG calls. Bit 0 served as the assignment bit. The remaining 300 bits were divided into two contiguous 150-bit physical streams. The assignment bit determined which stream received the Subject label and which received the PCS label.

Participants saw feedback based only on the Subject stream. The PCS was not displayed and served as the within-call reference. After every fifth experimental block, a separate 1,000-bit QRNG audit call was made without a target display, intention, or feedback. These calls provided an independent check on QRNG behavior across the study. They did not pass through the full experimental splitting and scoring pathway.

The pilot also included an AI condition as a pilot-specific procedural comparison. The AI condition was used to examine whether the task interface, timing structure, and intent-framed prompting produced patterns distinguishable from the Human and Baseline conditions. The AI condition is not a required component of the Paired-Delta Protocol.

These settings describe the pilot implementation and are not requirements of the general protocol. Future studies may use different stream lengths, partition rules, acquisition designs, QRNG providers, or metrics. These choices should be specified before data collection and calibrated for the intended analysis.

Figure 1. Paired-Delta QRNG architecture in Pilot

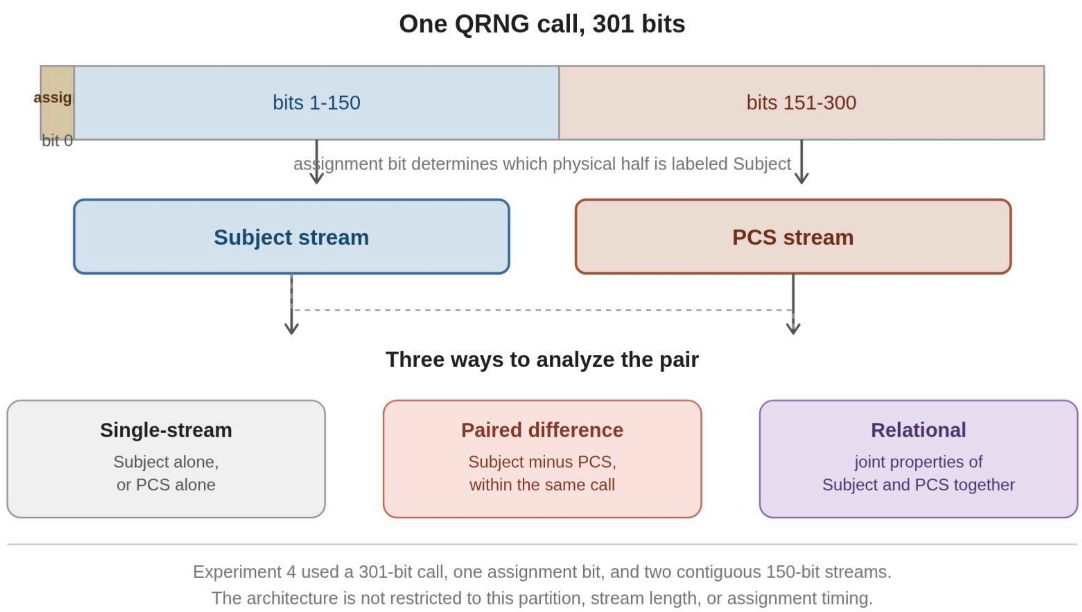

Figure 1. In the pilot a single QRNG call generates an ordered sequence that is divided into two physical streams. The implementation used a 301-bit call with one assignment bit and two contiguous 150-bit streams. The architecture supports Direct analysis of either stream, a within-call Paired-differential comparison, and Relational analysis of the two streams together.

Both hit rate and the single-scale H_{RS} score were available in the original experiment code. H_{RS} became the main worked example because it was sensitive to ordering within the short streams and produced a candidate pattern that warranted further examination in the exploratory analysis (Section 1.4).

The pilot used an empirical Baseline condition that consisted of automated blocks run on the same system without a Subject present. Baseline served as the between-condition empirical reference for Direct, Paired-differential, and Relational analyses.

1.4 Exploratory and Confirmatory Use

PDP can be used both when a researcher has a defined hypothesis and when the researcher is still exploring which features or measures are informative.

In a confirmatory study, the proposed effect, stream construction, metric, analysis family, comparison, calibration procedure, grouping structure, direction of the test, and decision rule are specified before the outcome data are examined. This use is appropriate when previous work or theory identifies a particular pattern for prospective testing.

In an exploratory study, the researcher may not yet know which metric, stream property, or relationship will be most informative. PDP can support a structured sweep across Direct, Paired-differential, and Relational analyses. This increases the opportunity to identify unusual patterns, so the scope of the search and its multiplicity must be reported transparently. Candidate patterns identified in this way require evaluation in new data under a prespecified analysis plan. The value of the exploratory mode is that it makes the search visible and helps define more focused questions for later testing.

The pilot used this exploratory mode. It was designed primarily to test the architecture across a broad set of measures rather than to provide a confirmatory test of an observer-related effect. It evaluated implementation, tested the behavior of paired subtraction under specified simulated disturbances, examined Direct, Paired-differential, and Relational measures, and identified calibration and interpretation problems that matter for future work.

The single-scale H_{RS} pattern is therefore presented as a worked example of the exploratory process. It emerged from the broader analysis and was subsequently examined using the Subject stream, PCS, and the paired difference. It is not presented as a confirmed Human-specific result.

For the prospective follow-up, the primary confirmatory test is the Direct comparison of Human and Baseline Subject-stream H_{RS} scores. Paired-differential and PCS analyses will serve as mandatory diagnostics because a condition-related change could affect both streams and be reduced by subtraction. These diagnostics will show how any Direct difference is distributed across Subject and PCS. They use the same underlying calls and are not independent confirmations.

Relational analysis remains exploratory in the current research program. It asks a different question about the relationship between Subject and PCS, and the pilot did not establish a sufficiently defined relational pattern for confirmatory testing.

1.5 The Three Analysis Families

PDP supports three analysis families that use the same paired acquisitions but describe different features of the data.

Direct analysis examines one stream at a time. A stream-level measure can be calculated for Subject or PCS and compared with Baseline. It asks whether a selected property of either stream differs across conditions.

Paired-differential analysis compares the two stream-level values within the same acquisition. For a selected measure M,

$$\Delta_M = M_{Subject} - M_{PCS}.$$

The paired difference asks whether Subject differs from its same-call PCS reference. Contributions expressed similarly in both stream statistics can be reduced by subtraction, while unequal contributions remain. The Subject and PCS values should therefore be examined alongside the paired difference when interpreting how a contrast is formed.

Relational analysis examines properties of the pair. It can measure association, covariance, temporal alignment, or the stability of the relationship between Subject and PCS. The central question is whether this joint behavior differs between the Human and automated Baseline conditions.

The relational case is worth stating in plain terms, because it is the least familiar of the three analyses and the easiest to conflate with a difference in level. Consider two dancers performing together. Being in step is not a property of either dancer alone, so a second dancer turns the pair itself into something to measure, and that pair-level property can be assessed in many ways. One, for example, may view how in step they are on average, whether that degree of synchrony holds steady or drifts, and whether their most extreme departures from the usual pattern happen together. None of these questions can be asked of a single dancer.

Two streams can each show ordinary marginal behavior and still differ in how they vary together across conditions. Their average paired difference can also remain near zero while their relationship differs. These joint properties require two streams and cannot be evaluated using a single retained stream.

Because all three analyses use overlapping data from the same calls, they should not be treated as statistically independent confirmations.

Figure 2. PDP Analysis taxonomy

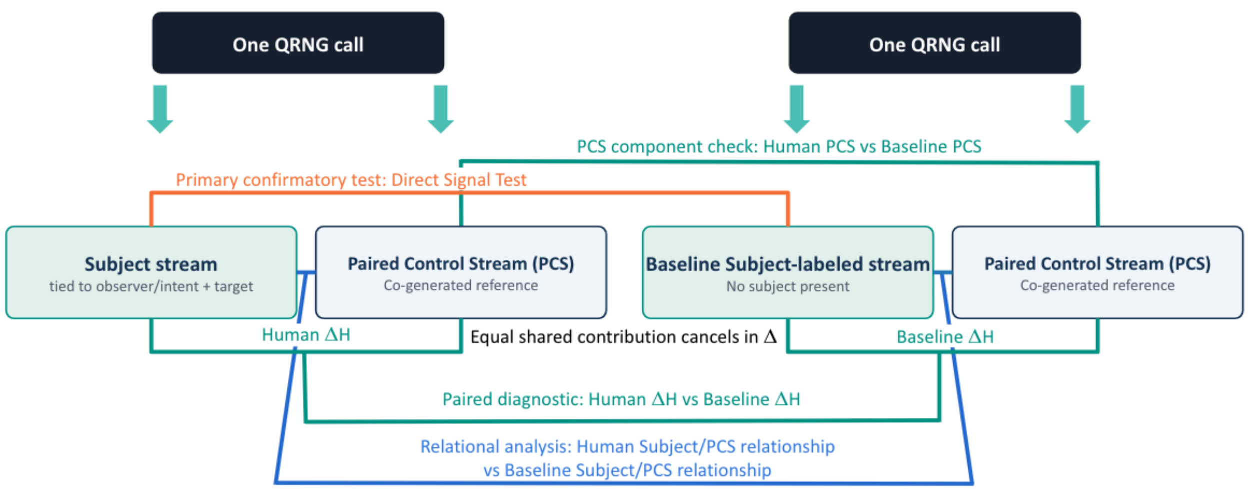

Figure 2. The same-call data support Direct, Paired-differential, and Relational analyses. Direct analysis examines Subject or PCS relative to Baseline. Paired-differential analysis compares Subject with PCS within the

same acquisition. Relational analysis examines properties of the pair. PCS is the within-call reference, while Baseline is the between-condition empirical reference.

2. Methods

2.1 Participants

A total of 121 human participants contributed data. Recruitment occurred through two main channels plus a small amount of personal outreach.

Most participants (approximately 100) were recruited through a survey-exchange platform where users complete other researchers' studies in exchange for credits to post their own. This group participated primarily for platform incentives and had no special reason to care about the topic beyond completing the task.

A smaller group was recruited through informal posts to Facebook and to the Reddit community r/remoteviewing. Facebook recruitment produced only a handful of participants, while the Reddit community provided most of the non–survey-exchange participants. Reddit participants were specifically recruited as people who believed in psychic phenomena and were asked to complete the study five to ten times.

Most participants therefore completed one session. Three participants completed five or more sessions. One was the principal investigator, who completed 18 sessions. Analyses of this subgroup are exploratory. Sensitivity checks examined recurrence across sessions, the influence of individual sessions, and the effect of excluding the principal investigator. These checks are reported in Sections 3.7.1 and 5.5.

Recruitment source, belief profile, and session count were confounded by design, so session-count differences cannot be interpreted as evidence of learning or a dose-response effect. This issue is examined further in Sections 3.6.2 and 5.4.

Intake data collected before each session included age and gender, ADHD and autism-spectrum diagnoses, attention-related self-reports, meditation frequency, openness to psi possibility, and current mental clarity. Meditation frequency and openness to psi possibility were analyzed as exploratory moderators of hit rate and H_{RS} score. The remaining intake measures were used for sample description or reserved for future analysis. These variables are included in the public dataset. Post-session ratings of focus, calm, and confidence, along with optional free-text feedback, were recorded but not analyzed.

2.2 Apparatus

Quantum random bits were generated using hardware quantum random number generators (QRNGs) accessed via public web APIs. Due to API rate limits and service availability constraints, three independent QRNG services were used sequentially during data collection: Outshift (Cisco Quantum Entropy Service), LFDR QRNG (Laboratory for Digitalisation Research; Germany), and the ANU

Quantum Random Numbers Server (Australian National University). The majority of blocks were collected using the first two providers.

Each experimental block fetched 301 bits in a single API call. Both the Subject stream and the Paired Control Stream (PCS) were derived from sequential halves of the same 301-bit call. This same-call origin allowed paired subtraction to remove contributions expressed equally in the two calculated stream statistics, but did not ensure that device characteristics affected the two halves or statistics equally.

Provider identity was not successfully stored at the block level because of a code error. It could be partly reconstructed from session and audit records, but remained incomplete and partly confounded with collection period and condition. Provider-related limitations are discussed further in Section 5.8.

The experiment ran in a web browser. Human participants saw a target color that remained constant for each five-block interval and was rerandomized after each audit call. The target pulsed at 5 Hz while the QRNG call was being generated. Target assignment and timing are described in Section 2.3.

2.3 Procedure

Each session was designed to contain 30 blocks. A target color was assigned for each five-block interval and rerandomized at every audit call. When ready, the participant pressed a button to initiate a single 301-bit QRNG call. During bit generation, the target pulsed at 5 Hz while the participant maintained focus.

Bit 0 served as the assignment bit, randomly determining which 150-bit half of the call was labeled the Subject stream and which was labeled the Paired Control Stream (PCS). Participants never saw the raw bitstream. After completion of the call, feedback was displayed as the hit rate of the Subject stream (proportion of bits matching the intended direction). The PCS was scored identically but not displayed.

Every fifth block, a separate 1000-bit audit call was made without task context (no target, no intention, no feedback). These audit calls provided separate monitoring measurements of QRNG behavior during data collection. Because they did not use the 301-bit trial-call processing path, they were analyzed separately from the experimental blocks.

2.4 Conditions

The pilot included Human, AI, and Baseline conditions. The Human and Baseline conditions provided the main experimental comparison. The AI condition was a pilot-specific procedural comparison and is not a required component of the Paired-Delta Protocol.

2.4.1 Human

Participants performed the intention task described in Section 2.3. They initiated each block, focused on the displayed target, and received feedback based on the Subject stream.

2.4.2 AI Agent

The experiment included an AI-driven condition implemented using GPT-4o-mini. This condition served two purposes. First, it provided a procedural control for task flow, timing, and interface dynamics under the paired-delta architecture. Second, it allowed an exploratory assessment of whether a non-biological agent, when given intent-framed prompts and feedback, would exhibit any measurable divergence from hardware baseline or any learning-like dynamics across blocks or sessions.

The AI condition consisted of two components: GPT-4o-mini, which generated text responses to prompts, and a Puppeteer automation layer that executed the corresponding browser actions. On each block, Puppeteer detected the "Ready" screen and provided GPT with a prompt mirroring what a human participant would see (e.g., "Focus intention on BLUE. Ready."). After GPT returned a response, Puppeteer initiated the QRNG call. During the "Fetching quantum data…" phase, GPT received a second prompt aligned with the pulsing target display. After each block, the agent received a reduced form of the feedback available to human participants (its hit-rate percentage for that block) before proceeding.

Despite this alignment, the AI condition differs from human participation in important ways. Human participants are self-paced and may delay initiation, adjust their internal state, or pause between blocks. GPT, by contrast, operates within the lifecycle of an API call and cannot intentionally delay or modulate timing beyond the duration of its response generation. Although the agent’s output gates the initiation of each QRNG call, it does not exercise open-ended control over when initiation occurs. This implementation represents the closest analogue achievable within a stateless API-driven framework, while acknowledging that it is not equivalent to human self-pacing.

In the present study, the AI condition therefore functions both as a procedural control and as an exploratory probe of whether intent-framed prompting in a non-biological system produces any detectable structure under the paired-delta architecture.

2.4.3 Baseline

The Baseline condition used fully automated execution without an agent present to receive or act on the intention framing. Block initiation and inter-block timing were scripted rather than self-paced.

Baseline provided the between-condition empirical reference for the direct, paired, and relational analyses. It was not an ideal theoretical reference and was not collected contemporaneously with the Human sessions. The implications of the different collection structures are discussed in Section 5.8.

2.5 Data Freeze

All data were frozen prior to analysis. SHA-256 fingerprints were computed for each file and stored in a manifest. All reported analyses use this frozen dataset.

The frozen block table contained 11,764 original rows. We first removed 100 blocks with missing required paired outcome or session metadata. The fewer-than-25-retained-block session rule then removed a further 57 blocks. The final trial dataset therefore comprised 11,607 blocks across 387 retained sessions (Human: 4,737 blocks across 158 sessions; AI: 3,780 across 126 sessions; Baseline: 3,090 across 103 sessions). Of these sessions, 386 contained 30 retained blocks and one Human session contained 27.

Separately, 1,891 audit calls were available across 389 sessions containing audit records. The 389 audit-containing sessions and the 387 retained trial-data sessions are different counts because the audit inventory was not restricted by the trial-data inclusion funnel.

2.6 Analysis Taxonomy and Stream-Level Metrics

The following metrics were prespecified and computed live for both Subject and PCS during data collection: hit z-score and p-value, cumulative range, single-scale H_{RS}, lag-1 autocorrelation, and block-level Shannon entropy. The single-scale H_{RS} score became the main worked example because it was sensitive to ordering within the short streams and produced a candidate pattern that warranted closer examination (Section 1.4). Additional ordering and complexity measures were later reconstructed from the frozen raw bits as part of the exploratory analysis. These included run count, run-length entropy, higher-order n-gram entropy, and mutual information. Run count, lag-1 autocorrelation, and run-length entropy proved mechanically redundant and were treated as non-independent measures.

2.6.1 Analysis Taxonomy

As introduced in Sections 1.2–1.5, PDP organizes analyses by analysis family and by the property being measured. Table 1 shows examples of the questions each combination can support. Researchers can select the analyses that fit their study.

For any selected stream-level measure M, Direct analysis can compare the Subject stream across conditions:

$$\bar{M}_{Subject,Human} - \bar{M}_{Subject,\,Baseline}.$$

Paired-differential analysis first calculates the within-call difference:

$$\Delta M \;\; = M_{Subject} - M_{PCS}$$

which can then be compared across conditions:

$$\Delta\bar{M}_{Human} - \Delta\bar{M}_{Baseline}$$

These expressions are metric-agnostic. M can represent any appropriately specified stream-level measure.

Table 1. Two-Dimensional Analysis Taxonomy for the Paired-Delta Protocol

	Direct analysis	**Paired differential analysis**	**Relational analysis**
What the analysis family	Conventional stream-level	Within-call internal reference and	Coupling, alignment,

adds	measurement	conditional common-mode rejection	stability, lag structure, and joint shape
Location	Mean level in Subject or PCS	Mean Subject-minus-PCS difference	Average association or coupling
Dispersion	Variability within one stream	Variability of the paired difference	Variability of contemporaneous or lagged coupling
Composition	Zero/one balance in one stream	Difference in bit balance between Subject and PCS	Joint bit combinations or same-position co-occurrence
Sequential Structure	Persistence, runs, autocorrelation, n-gram patterns, ordering complexity	Difference in the same ordering measure	Local, lagged, or directional cross-stream dependence
Spectral structure	Periodicity, oscillations, frequency power, spectral entropy	Between-stream difference in spectral features	Cross-spectrum, coherence, phase, shared periodicity
Trajectory	Change across blocks or sessions	Change in the paired difference	Change in coupling over time
Joint-distribution Shape	Not applicable	Not applicable	Nonlinear dependence, tails, clusters, copula shape

Table 1. The columns show the analysis family and the rows show the property being measured. The entries are examples of possible questions, and not every analysis was used in the pilot. Some analyses span more than one property. For example, the windowed relational-correlation-variance analysis in Section 3.7.2 is

listed under Dispersion because it measures variability in correlation. Because it is calculated across temporal windows, it also relates to Trajectory.

The taxonomy can be applied either prospectively or exploratorily, as described in Section 1.4.

2.6.2 Hit rate

Hit rate was the proportion of bits matching the intended direction, calculated separately for Subject and PCS.

$$\Delta_{rate} = rate_{Subject} - rate_{PCS}$$

As with other paired-delta comparisons (Section 1.2), this subtraction cancels contributions expressed equally in both streams' hit rates while retaining contributions that affect them unequally. A surviving difference indicates asymmetry between the labeled streams. It does not by itself indicate deviation from 0.5.

2.6.3 Single-Scale H_{RS} Ordering Score

The single-scale H_{RS} ordering score was implemented in the experiment's JavaScript source as `hurstApprox` before the pilot data were analyzed. It was used as an exploratory, ordering-sensitive screening measure.

Rescaled-range analysis is classically used to estimate the Hurst exponent from how R/S changes across multiple sequence lengths. Because the implemented function evaluates only the complete 150-bit block, it does not provide a multi-scale estimate of the Hurst exponent. Instead, it produces a single-scale score, denoted H_{RS}, which was used here as an exploratory measure of within-block ordering.

Raw bits were first mapped to -1 and +1. For each 150-bit stream, each value's deviation from the stream mean was accumulated in sequence. R was the difference between the highest and lowest points reached by this cumulative-deviation trajectory, and S was the standard deviation of the 150 mapped values, calculated using n in the denominator. The score, denoted H_{RS}, was calculated as

$$H_{RS} = \frac{\log(R/S)}{\log(n)},\ n\ =\ 150$$

Values were constrained to $[0, 1]$, consistent with the limits of the Hurst exponent that the function was originally intended to approximate.

R is the range of the cumulative-deviation trajectory, not the range of the raw bit values. Because this trajectory depends on the order in which the bits occur, rearranging the same 150 values generally changes R and therefore H_{RS}. By contrast, S and hit rate remain unchanged when the same bits are rearranged. The score can therefore detect aspects of ordering that are not captured by hit rate alone, although changing the overall bit composition may also affect the score.

Higher H_{RS} values indicate more extended cumulative excursions, while lower values indicate more constrained excursions. Different sequences can produce the same value, so the score does not uniquely identify the ordering pattern responsible for a result. Because it is calculated at only one sequence length, it is not interpreted as a Hurst exponent or as evidence of long-range dependence. The shuffle analysis in Section 3.6.3 and the simulations in Section 5.10 characterize its response under the conditions examined.

2.7 Statistical Framework

Hit rates and single-scale H_{RS} scores have different distributional properties and were analyzed accordingly.

2.7.1 Hit rate analysis

The Bayesian analysis modeled each block’s paired target-aligned hit-rate difference, calculated as Subject minus PCS, as a continuous outcome. Fixed effects were included for Human 5+ and AI relative to Baseline and for centered within-session block position. Random intercepts for database identifier and session represented the grouping structure available in the stored data. The 20 database identifiers for the Human 5+ subgroup included Human contributors and accounts used for the automated AI and Baseline conditions; they do not represent 20 independent contributors or fully encode the automated collection-burst structure.

Weakly informative priors were used for all model parameters. The model was implemented in PyMC and sampled using NUTS with four chains, 2,000 tuning draws, and 2,000 retained draws per chain. Results are reported as posterior means and 95% highest-density intervals in Section 3.5.

Hit rate was included primarily to give participants a concrete task during each session and to confirm that block generation and scoring were functioning correctly. The pilot was not large enough to reliably detect the very small hit-rate effects reported in this literature. The weighted mean deviation from chance for the original Radin and Nelson (1989) meta-analysis is 0.00018, as reported in Bösch, Steinkamp, and Boller (2006, p. 501). A post hoc sensitivity calculation indicated that detecting an effect of this size would require approximately 51,320 participants for 80% power or 68,702 participants for 90% power. This calculation was used only to place the precision of the hit-rate results in context and was not part of the Bayesian model.

2.7.2 Paired Single-Scale H_{RS} score difference (ΔH_{RS})

The single-scale H_{RS} score has substantial finite-sample variability at $n = 150$.

Primary uncertainty analyses preserved the observed hierarchy. Human contributors were resampled first, followed by whole sessions within contributors. Baseline sessions were resampled as whole units, conditional on the two machines and collection periods represented in the observed Baseline data.

Sensitivity analyses compared block-weighted and equal-contributor/equal-session estimates. Equal-session weighting was also calculated, but it was identical to block weighting because every session in the Human 5+ vs Baseline comparison contained 30 blocks. Whole-session recurrence and leave-one-session-out analyses tested whether the Human 5+ pattern was concentrated in particular sessions.

A composition-conditional order-randomization analysis calibrated selection of the largest positive contrast across the documented 2+, 3+, and 5+ thresholds while preserving each stream's observed bit composition. Block-level KS and t-tests are retained only as legacy descriptive calculations and are not used for participant- or population-level inference.

Paired comparisons were computed as:

$$\Delta H_{RS} = H_{RS,\,Subject} - H_{RS,PCS}$$

The analysis did not apply an Anis–Lloyd (Anis & Lloyd, 1976) correction to individual scores. At a fixed sequence length, this correction adds the same value to both 150-bit scores and therefore cancels exactly in their paired difference. This does not show that the estimator behaves the same when Subject and PCS have different underlying ordering structures. That issue is examined in the simulations in Section 5.10.

2.7.3 Relational analysis framework

Relational analyses examined the association between the Subject and PCS streams.

The pooled relational battery included Pearson correlation between Subject and PCS stream-level scores and same-position mutual information between the two 150-bit sequences. The mutual-information analysis used a circular-shift comparison. The PCS sequence was rotated through its nonzero circular offsets before mutual information was recalculated. This preserved each stream's complete sequence while breaking the recorded same-position alignment between the streams.

Additional analyses examined whether the relationship between the streams varied across local windows, sessions, conditions, or specified temporal offsets. These analyses asked whether the relationship was stable, rather than only whether the average relationship differed.

The grouping structure used for relational analyses followed the same logic as the other analyses in this paper. Human relational statistics were resampled by contributor because contributors could participate repeatedly. Baseline and AI relational statistics were resampled by session because neither condition contained a comparable repeated-contributor structure. A sensitivity check that grouped Baseline and AI sessions by continuous collection bursts did not indicate that session-level resampling understated uncertainty.

The windowed-correlation-variance and lagged-coupling analyses in Section 3.7.2 were added after the pooled relational battery was complete. They are therefore reported as exploratory extensions rather than prespecified tests.

2.7.4 Interpretation of p-values

The subgroup, sensitivity, and additional pattern analyses were exploratory. Their p-values describe compatibility with the relevant null model and are interpreted alongside effect size, uncertainty, and the exploratory search.

The 2+, 3+, and 5+ groups were nested and therefore did not represent independent comparisons. Their block-level KS results are reported descriptively. A composition-conditional order-randomization analysis evaluated both the fixed 5+ contrast and the largest positive contrast across the three documented thresholds while preserving each stream’s observed bit composition. This analysis addresses selection among those three thresholds only. It does not adjust for the wider historical exploratory search. Confirmatory conclusions require a preregistered replication.

2.8 Unit of Analysis

Block-level analyses describe structural differences within the observed data. QRNG calls were generated independently, and the task contains no adaptive stimulus-response channel. However, blocks contributed by the same participant cannot be treated as statistically independent observations for participant- or population-level inference. This dependence is reflected in the clustered bootstrap reported in Section 3.7.1, which produces wider uncertainty intervals.

The within-call Subject-minus-PCS comparison is defined at the level of a single QRNG call. It therefore does not depend on clustering across participants or sessions. Clustering becomes relevant when paired scores are compared across conditions or aggregated across repeated observations from the same contributor. Those analyses must preserve the participant and session structure.

For the Human 5+ analysis, the estimand is the difference between Human 5+ and Baseline in the mean within-block paired score, $\Delta H_{RS} = H_{RS,Subject} - H_{RS,PCS}$. Results were summarized using block-weighted and equal-contributor/equal-session estimates. Equal-session weighting gave the same result as block weighting because every session included in the Human 5+–Baseline comparison contained 30 retained blocks.

The Human 5+ subgroup contains 840 blocks organized within 28 sessions from three retrospectively selected contributors. The blocks permit detailed description of the observed sequences, but they are not 840 independent observations for participant- or population-level inference. Hierarchy-preserving bootstrap intervals and session-level influence analyses retain the observed contributor and session structure.

Accordingly, the Human 5+ estimates describe the observed pattern and its sensitivity to weighting within these three contributors. They do not estimate population variability or establish generalizability to new participants. Whole-session recurrence, leave-one-whole-session-out influence, and a numerical pooled-versus-leave-PI-out comparison are reported in Section 3.7.1.

2.9 Exploratory Human 5+ Session Subgroup

The Human 5+ subgroup comprised the three participants who completed at least five sessions, contributing 28 sessions and approximately 840 blocks. The threshold was examined retrospectively and was initially motivated by the repeated-session recruitment described in Section 2.1. The additional 2+ and 3+ session-count thresholds were examined as follow-up comparisons and are reported in Section 3.6.4.2.

The Human 5+ pattern was summarized using block-weighted and equal-contributor/equal-session estimates. Equal-session weighting was identical to block weighting because every session included in the Human 5+–Baseline comparison contained 30 retained blocks. The equal-contributor analysis tested whether the result changed when each contributor received equal weight. Recruitment and interpretive limitations are discussed in Sections 5.2 and 5.5.

2.10 Computational Environment

Analyses were conducted in Python 3.12.13 using NumPy 2.0.2, pandas 2.2.2, and SciPy 1.16.3. The full analysis pipeline and frozen dataset are available in the public repository (link provided in Data Availability). The analyses extracted hit counts and raw bit arrays from the frozen dataset. Structural measures were recomputed from the raw bits rather than taken from the application's stored summary fields. These included within-block lag-1 autocorrelation and the cumulative-deviation range used to calculate the single-scale H_{RS} score. Lag-1 autocorrelation was analyzed directly. Cumulative range was used only as a component of the H_{RS} score. The original application variables are documented in the repository variable schema.

2.11 Preregistered Synthetic Diagnostic-Recovery Validation

After the PDP diagnostic rules were developed, a separate preregistered held-out synthetic validation tested whether the frozen rules could recover deliberately constructed marginal and relational patterns. The retained Study 1 Baseline cohort was used to build the controlled test datasets. The validation datasets were generated only after the diagnostic rules had been frozen. The generating-condition key was withheld during analysis, and classifications were made from blinded feature tables before being compared with the known conditions.

The validation tested three prespecified synthetic pattern classes. First, marginal and zero-lag relational alterations were generated separately to test whether the framework could distinguish a stream-level change from a change in the relationship between streams. Second, relational dependence was introduced at temporal offsets from −2 through +2 to test lag localization. Third, coupling instability was generated by varying the local

Subject–PCS relationship across nonoverlapping 10-block windows while keeping average coupling near zero. Null datasets served as negative controls.

The frozen rules used fixed thresholds for each pattern class. A marginal alteration was identified when the paired-H_{RS} shift exceeded 0.004 without a zero-lag relational flag. A zero-lag relational alteration was identified when the absolute Subject–PCS H_{RS} correlation exceeded 0.15. Temporal dependence was localized using the largest mean within-session correlation across lags −2 through +2. Coupling instability was identified when the variance of 10-block local correlations exceeded 0.50 while the absolute mean local correlation remained below 0.05.

The analyses used to develop these rules were completed before preregistration and were not counted as prospective validation evidence. The preregistration did not specify a threshold for declaring absence of lagged dependence in null data, so none was added after the held-out results were generated.

3. Results

3.1 PCS Calibration Under the Observed Conditions

The Paired Control Stream (PCS) serves as the within-call reference for the experimental design. Across all conditions, its hit rate was 0.500253 (session-bootstrap $95\%\ CI\ [0.499504,\ 0.501022]$; sign-flip $p\ =\ 0.5160$).

No condition-related difference was detected in mean PCS hit rate $(p\ =.7769)$ or mean single-scale H_{RS} score $(p\ =.4696)$. Additional hierarchy-aware checks found no clear evidence of across-block lag-1 dependence, unusual run structure, within-session hit-rate drift, or abnormal hit-count dispersion. These checks describe how the PCS behaved under the observed study conditions. Figure 3 shows the distributions and session-level estimates.

Figure 3. PCS measures across conditions.

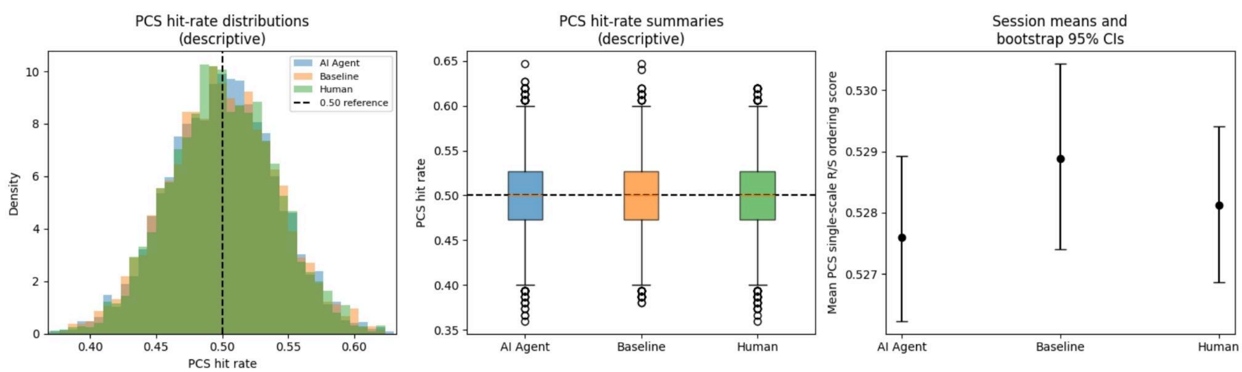

Figure 3. The first two panels show the observed block-level PCS hit-rate distributions and summaries across Human, AI, and Baseline conditions.

The third panel shows mean session-level PCS single-scale H_{RS} scores with bootstrap 95% confidence intervals. No condition difference was detected for mean PCS hit rate ($p = .7769$) or mean PCS H_{RS} score ($p = .4696$). These analyses test for detectable differences. They do not establish exact equivalence or show that the PCS is random.

These analyses evaluate the PCS alone. We therefore examined the relationship between the two streams directly, using within-call and cross-call comparisons.

3.2 Relational Structure and Paired-Difference Behavior Under Baseline

A concern with the paired-delta design is that the two streams from the same QRNG call could share structure. PCS alone cannot show how the paired difference behaves (Section 3.1). We therefore examined selected properties of the Subject–PCS relationship using Baseline blocks only ($n = 3,090$), where no Human or AI intention task was applied. Section 3.2.5 extends these analyses across all three conditions.

3.2.1 Common-mode correlation

Shared linear structure in consecutive halves of a single call could produce a positive association between the Subject and PCS scores across Baseline blocks. No such association was detected at the resolution of these analyses. For the single-scale H_{RS} score, Pearson $r = + 0.0037$, with a whole-session bootstrap 95% confidence interval of $[-\ 0.0306, +\ 0.0371]$ and a within-session permutation $p\ =.8357$. The corresponding Spearman correlation was $\rho\ =+\ 0.0027$ ($p\ =.8791$). For target-aligned hit rate, Pearson $r\ =+\ 0.0149$, with a 95% confidence interval of $[-\ 0.0248, +\ 0.0551]$ and a within-session permutation $p\ =.3966$. No same-call linear or rank association was detected in the 3,090 Baseline blocks. Weaker, nonlinear, cross-lagged, or time-varying relationships may still be present.

3.2.2 Empirical Baseline centering and physical-half symmetry

Across the 3,090 Baseline blocks from 103 sessions, the mean paired single-scale H_{RS} score difference was −0.0017, with a whole-session bootstrap 95% confidence interval of $[-\ 0.0039, +\ 0.0004]$. The observed distribution had little skew $(-\ 0.05)$ or excess kurtosis $(-\ 0.11)$. The mean paired target-aligned hit-rate difference was $+\ 0.0002$, with a whole-session bootstrap 95% confidence interval of $[-\ 0.0018, +\ 0.0022]$. Thus, neither analysis detected a systematic Baseline difference between the Subject and PCS streams.

Because the implemented protocol divides each call into two sequential 150-bit physical halves before assigning the Subject and PCS labels, the physical positions were also compared directly. The mean physical A-minus-B H_{RS}-score difference was −0.00068, with a whole-session bootstrap 95% confidence interval of $[-\ 0.00301, +\ 0.00172]$. No systematic physical-half asymmetry was detected at the implemented sequence length. These results do not rule out a boundary or position effect.

3.2.3 Same-call alignment versus within-session cross-call re-pairing

To examine whether the actual within-call alignment contributed detectable linear covariance, each observed Subject score was compared with its PCS score from the same call and with PCS scores reassigned from other blocks within the same session. Reassigning PCS blocks preserved the session membership and the separate Subject and PCS distributions while breaking their original call-level alignment. The analysis used 10,000 within-session permutations.

The comparison was expressed through the variance of the paired difference. For the single-scale H_{RS} score, the observed paired-delta variance was 0.9963 times the sum of the separate Subject and PCS variances. This ratio fell within the 95% permutation reference interval of $[0.9624, 1.0321]$. For target-aligned hit rate, the corresponding ratio was 0.9851, within a reference interval of $[0.9653, 1.0341]$. Thus, preserving the actual within-call alignment did not produce a resolved covariance contribution relative to within-session cross-call re-pairing.

This analysis is not independent of the Pearson correlation in Section 3.2.1. Paired-difference variance depends on the two stream variances and their covariance, so both analyses reflect the same underlying covariance information.

3.2.4 Interpretation

The Baseline analyses found no clear same-call relationship at the available resolution. The correlation and re-pairing analyses describe the same underlying covariance relationship in different ways, while the additional dependence checks examined other forms of joint structure. These results apply only to the relationships tested under the observed Baseline conditions and do not establish that the streams were completely independent.

These analyses illustrate the Relational family introduced in Section 1.5 by examining Subject and PCS jointly. Additional relational and dependence checks are reported in Section 3.2.5.

The separate finite-sample behavior of the single-scale H_{RS} score is examined in Section 5.10.

3.2.5 Additional structure and dependence checks

Additional analyses examined lag-1 autocorrelation, recurring bit patterns, entropy, and several aspects of the relationship between the Subject and PCS streams. The temporal-structure analyses found no clear condition differences in the features tested.

Mutual information calculated directly from same-position raw bits averaged approximately 0.0047 bits. Same-position mutual information, evaluated against the circular-shift reference described in Section 2.7.3, showed a resolved negative excess for Human at the contributor level ($mean =- 0.000262$, $95\% \, CI \, [- 0.000438, - 0.000078]$, $p = 0.0058$, $n = 121$ contributors). Baseline and AI estimates were smaller and individually unresolved: Baseline mean $=- 0.000143, \; 95\% \, CI \, [- 0.000365, + 0.000083], \; p =.215, \; n = 103$ sessions; AI mean$=- 0.000134, \; 95\% \, CI \, [- 0.000330, + 0.000066], \; p =.189, \; n = 126$ sessions. Direct contrasts did not resolve a difference between Human and Baseline ($p = 0.98$) or between AI and Baseline ($p = 0.997$). Because the condition contrasts were unresolved, this pattern does not establish a Human-specific mechanism and is treated as an open, low-priority relational observation.

Using contributor-clustered bootstrap for Human and session-clustered bootstrap for Baseline and AI, the condition-comparison analyses found no resolved differences in Subject–PCS correlation, the variance of the paired relationship, or the joint two-dimensional distribution. Fisher-z correlation contrasts were $+ 0.0072, \; 95\% \, CI \, [- 0.0397, + 0.0519]$, for Human minus Baseline; $+ 0.0061, \; 95\% \, CI \, [- 0.0394, + 0.0508]$, for AI minus Baseline; and $+ 0.0011, \; 95\% \, CI \, [- 0.0424, + 0.0431]$, for Human minus AI.

All three pairwise variance ratios were between 0.927 and 1.055, and all energy-distance comparisons had $p \geq 0.19$. An extended battery of lag-1 autocorrelation, n-gram entropy, and entropy-rate slope, computed in direct and relational forms across all three conditions, produced unresolved contrasts in 47 of 48 comparisons. The remaining nominal result sat at the boundary of its interval and was consistent with chance within an unadjusted 48-comparison family. The analyses therefore provide no clear support that the broader Subject–PCS relationship differed between conditions at the resolutions tested.

3.3 Artifact Injection Test

Because the observed Baseline data did not contain a disturbance of known type and magnitude, we tested the protocol using synthetic artifact injections applied to the raw bits from the 3,090 retained Baseline calls (Section 3.2). Before injection, the raw physical halves were reconstructed and passed through the recorded Subject–PCS assignment and scoring pipeline. All reconstructed scores matched the frozen data. These simulations tested whether the pipeline behaved as expected under specified disturbances. They did not test whether PDP removes naturally occurring artifacts or whether cancellation operates at the resolution required for effects on the scale reported in the observer-RNG literature.

Two synthetic disturbances were introduced before Subject–PCS assignment. One altered literal-bit composition and the other increased within-stream persistence. Each was tested across several strengths under synchronized and independent-position implementations. The exploratory validation used 50 realizations at each nonzero level. A subsequent preregistered analysis repeated the full grid with 100 realizations and a fixed cancellation criterion, described in Section 3.3.4.

After each injection, the complete assignment and scoring pipeline was rerun. Changes in Subject, PCS, and their paired difference were measured relative to the original values. Literal-bit results were also examined separately for target-0 and target-1 blocks because a literal shift can have opposite target-aligned consequences.

Two additional controls tested cases in which cancellation should not occur. A target-relative intervention applied only to Subject tested whether the paired difference retained a stream-specific change. An unequal-half persistence condition tested whether it retained a disturbance affecting the two physical halves at different strengths. Together, these controls tested both expected cancellation and expected failure to cancel.

3.3.1 Shared literal-bit composition bias

For each Baseline call, a randomly selected direction toward literal 0 or literal 1 was applied to both physical halves before Subject–PCS assignment. At each dose (0.02, 0.05, 0.10, and 0.20), the dose represented the probability that a bit position would be replaced by the selected literal value. Each nonzero dose was evaluated across 50 independently generated artifact realizations.

Two forms of the disturbance were tested. In the synchronized-mask condition, the same positions were selected in both halves. In the independent-position condition, each half received the same direction and dose but at different positions. Because literal-bit shifts have opposite target-aligned consequences for target-0 and target-1 blocks, results were also examined separately by target.

Across both implementations, the mean change in the paired hit-rate difference remained close to zero at every dose. At the largest dose of 0.20, the mean recovery error was $(-\ 0.000007)$ for synchronized masks and $(-\ 0.000021)$ for independent-position masks; both whole-session bootstrap intervals included zero. The block-level recovery-error RMS nevertheless increased with dose, reaching 0.0257 and 0.0347, respectively. Thus, the shared disturbance produced little systematic shift in the paired difference when averaged across calls, while still adding block-level variability. Independent affected positions produced more residual variability than synchronized positions because equal artifact strength does not guarantee identical changes in the two streams.

3.3.2 Shared persistence disturbance

Increased within-stream persistence was introduced at the raw-bit level by giving selected positions a specified probability of repeating the preceding bit. The disturbance was applied to both physical halves before Subject–PCS assignment, after which the single-scale H_{RS} scores and paired differences were recomputed. Doses of 0.05, 0.10, 0.20, and 0.35 were each evaluated across 50 independently generated artifact realizations.

The synchronized-mask and independent-position implementations described in Section 3.3.1 were used here as well. Across both implementations, the average change in the paired H_{RS} score difference remained close to zero. At the largest dose of 0.35, the mean recovery error was +0.00109 for synchronized masks and +0.00111 for independent-position masks, with both whole-session bootstrap intervals including zero. Block-level recovery-error RMS increased with dose, reaching 0.0687 and 0.0697.

Equal persistence strength therefore produced little systematic displacement of the average paired difference while still increasing block-level variability. These results apply only to the persistence mechanisms tested.

3.3.3 Stream-specific and unequal-half controls

A labeled Subject-only intervention served as an implementation check for the assignment and scoring pipeline. Selected Subject-stream bits were replaced with the block's actual target value while PCS was left unchanged. Replacement probabilities of 0.02, 0.05, 0.10, and 0.20 produced mean increases in the paired difference of 0.0100, 0.0249, 0.0500, and 0.1000, the expected increase of approximately one-half the replacement probability. This verified that a labeled Subject-only change passed through the implemented assignment, target-scoring, and subtraction pipeline as specified.

A separate unequal-half control applied persistence to physical half A at twice the dose applied to physical half B. Unlike the equal-strength shared-persistence simulations, this disturbance produced a small systematic residual in the paired single-scale H_{RS} score difference. At doses of 0.10, 0.20, and 0.35 for half A, the mean recovery errors were +0.00069, +0.00106, and +0.00181, respectively, with whole-session bootstrap intervals excluding zero. This shows that paired subtraction need not cancel disturbances that affect the two physical halves unequally.

3.3.4 Interpretation

The artifact injections showed the expected operating pattern. Shared disturbances produced little systematic displacement of the paired difference on average, while the Subject-only and unequal-half controls remained detectable. Block-level variability could still increase even when the average paired difference remained near zero. These results describe the controlled mechanisms and doses tested here.

Pairing also carries a known precision cost. When the two stream statistics have similar variances and are approximately uncorrelated, their difference has about $(\sqrt{2})$ times the standard deviation of either stream alone. PDP therefore trades some statistical precision for reduced systematic displacement when a contribution affects both selected stream statistics similarly. Additional data can reduce sampling uncertainty, whereas a systematic common-mode contribution in a single-stream analysis does not disappear simply because more observations are collected.

The injection analysis was subsequently formalized in a preregistration filed August 19, 2026 (OSF, DOI 10.17605/OSF.IO/BG9YS), fixing an equivalence bound of

$\delta = 0.002$, derived from the resolution floor of a whole-session cluster bootstrap on the real unperturbed Baseline series, and a locked realization count of $R = 100$. The grid comprised 16 injection conditions evaluated on both hit rate and H_{RS}, producing 32 outcome cells. Four cells had already been computed before registration and were disclosed under Prior Knowledge. The remaining 28 were prospective. Under the preregistered classification rule, 30 of the 32 cells had confidence intervals entirely within $\pm\delta$. The two exceptions were the persistence dose 0.35 H_{RS} cells. The independent-position cell produced a mean recovery error of $+ 0.00091,\ 95\%\ CI\ [- 0.00052,\ + 0.00222]$, and the synchronized cell produced $+ 0.00113,\ 95\%\ CI\ [- 0.00026,\ + 0.00242]$. Both intervals included zero but extended beyond the equivalence bound and were therefore classified as indeterminate rather than successful or failed cancellation.

The width of the two indeterminate intervals was driven mainly by the 103 observed Baseline sessions rather than by the number of simulated realizations. Diagnostic work indicated that precision had approached a floor by about $R = 50$, so $R = 100$ was selected on precision-per-compute grounds. The diagnostic work had already revealed the four outcomes disclosed under Prior Knowledge. The connection between the realization-count diagnostic and the known outcomes was not fully described in the registration and is disclosed here.

3.3.5 Real-World Perturbation: QRNG Provider-Identity Validation

A dedicated follow-up study directly tested provider-dependent disturbance using a real, controlled perturbation rather than simulation: 269 machine-only sittings (8,042 blocks, no human or AI involvement) with QRNG provider identity (Outshift vs. LFDR) explicitly controlled and logged per block, balanced 15/15 within each sitting. A whole-sitting bootstrap on the design's own unperturbed paired difference gave a resolution floor of delta ≈ 0.0014. Neither an unpaired provider comparison nor a within-sitting paired-by-provider comparison detected a significant hit-rate or H_{RS} difference between providers, and both ruled out an effect near the scale of this study's own benchmark effect sizes (≥0.003) on the upper side; neither, however, achieved a full equivalence certification at delta, so the result is best characterized as no detected provider effect at a resolution the design cannot yet fully certify, rather than confirmed equivalence.

3.4 QRNG Audit-Sequence Checks

Separate 1,000-bit QRNG audit sequences were scheduled after blocks 5, 10, 15, 20, and 25. Based on completed block records, 1,945 audit opportunities occurred across 390 sessions that reached at least one scheduled audit point. The frozen audit table contains 1,891 audits across 389 sessions; 54 scheduled audits were not recorded. Missing audits were concentrated in the AI condition (47 of 636 opportunities), compared with Human (6 of 794) and Baseline (1 of 515), and all occurred during sessions whose recorded block-level provider was `qrng-race`.

Because condition, provider, collection date, and execution regime were partly confounded, the source of this missingness cannot be isolated. Audit availability was therefore not treated as evidence that monitoring coverage was equivalent across conditions.

Of the 1,891 recorded audits, 120 lacked complete stored results for the three-test battery. The Frequency, Runs, and Longest Run of Ones subtests were reconstructed from the underlying 1,000-bit audit sequences and validated against the 1,771 audits with complete stored results. The reconstructed classifications matched the stored pass/fail results in every case. The implemented Runs calculation used a denominator different from the standard NIST SP 800-22 formula, so these audits should be interpreted as results from the implemented three-subtest battery rather than as validation against the full NIST suite.

Among the 1,891 audits, 1,845 passed the complete battery and 46 failed at least one subtest, giving an all-three pass rate of 97.57% (session-cluster bootstrap $95\%\ CI:\ 96.89\%–98.20\%$). The 46 failures occurred in 46 different sessions.

Table 2. Three-Subtest Audit Results

Measure	Human	AI Agent	Baseline	Overall

Audits	788	589	514	1,891
All-three pass rate	97.84%	97.62%	97.08%	97.57%
Frequency pass rate	98.22%	98.81%	97.86%	98.31%
Runs pass rate	100.00%	100.00%	100.00%	100.00%
Longest Run pass rate	99.11%	98.30%	98.83%	98.78%

Table 2. Whole-session permutation comparisons found no condition difference in complete audit-battery failure rates after Holm adjustment, with all adjusted $p = 1.00$. This comparison does not account for the condition-associated difference in audit availability.

Failure rates were also summarized descriptively by QRNG provider and collection month. No inferential conclusions were drawn from these summaries because provider, date, and collection regime were partly confounded, and some provider samples were very small.

3.5 Mean-Level Hit-Rate Results

The primary hit-rate analyses compared the Human Subject-minus-PCS difference with Baseline. AI provided a secondary comparison. The retrospectively defined Human 5+ subgroup is included descriptively.

Table 3. Target-aligned hit-rate estimates by condition

Condition	Subject mean, % [95% interval]	PCS mean, % [95% interval]	Subject−PCS, percentage points [95% interval]	Difference from Baseline, percentage points [95% interval]
Baseline	50.0179 [49.8831, 50.1575]	49.9978 [49.8434, 50.1443]	+0.0201 [−0.1743, +0.2265]	Reference
AI Agent	50.0284 [49.9049, 50.1506]	50.0060 [49.8813, 50.1365]	+0.0224 [−0.1328, +0.1758]	+0.0023 [−0.2532, +0.2524]
Human-all participants	50.0386 [49.9096, 50.1532]	50.0585 [49.9183, 50.1898]	−0.0200 [−0.2034, +0.1621]	−0.0400 [−0.3190, +0.2246]
Human 5+ exploratory subgroup	50.2238 [49.9841, 50.5911]	50.1365 [49.9035, 50.5141]	+0.0873 [−0.3674, +0.5875]	+0.0672 [−0.4159, +0.6052]

Table 3. Subject and PCS means are shown separately, followed by the within-call paired difference and its contrast with Baseline. All intervals for the paired differences and Baseline contrasts included zero. No resolved difference was detected for AI, Human-all, or the exploratory Human 5+ subgroup.

A complementary Bayesian model specifically compared Human 5+, AI, and Baseline. Its estimated paired differences were similarly close to zero: +0.018 percentage points for Baseline $(95\%\ HDI:\ -0.275\ to\ +0.283)$, +0.019 for AI $(-0.196\ to\ +0.221)$, and +0.060 for Human 5+ $(-0.372\ to\ +0.503)$. The estimated Human 5+ versus AI contrast was +0.041 percentage points $(-0.451\ to\ +0.510)$.

The pilot therefore showed no clear mean-level hit-rate difference between conditions. A session-clustered sensitivity analysis indicated that the Human-all versus Baseline comparison had approximately 80% power to detect a difference of 0.372 percentage points under the specified location-shift model.

3.6 Exploratory Ordering Analysis: Single-Scale H_{RS} Score

The single-scale H_{RS} score was used to examine ordering within the paired streams. The analysis first considered the full Human cohort and then explored session-count subgroups, including the retrospectively identified Human 5+ subgroup.

3.6.1 Full Human cohort

The mean paired H_{RS} score difference was +0.00113 in the full Human cohort and −0.00174 in Baseline, producing a Human-minus-Baseline contrast of +0.00287. The hierarchy-preserving 95% interval was approximately -0.00001 to +0.00575 and narrowly included zero.

Unclustered block-level analyses produced $t = 1.94,\ p = .0521$, for the mean comparison and $D = 0.0226,\ p = .2884$ for the KS comparison. These are reported descriptively because they do not account for repeated observations within participants and sessions.

The full-cohort analysis therefore did not resolve a Human-Baseline difference.

3.6.2 Session-Count Pattern (Exploratory)

The paired single-scale H_{RS} score varied descriptively across the non-overlapping Human session-count strata.

Table 4. Mean paired ΔH_{RS} relative to the paired-null reference and empirical Baseline

Condition or Group	**Blocks**	**From paired-null reference**	**From empirical Baseline**
Baseline	3,090	−0.00174	Reference

AI Agent	3,780	+0.00094	+0.00268
Human-all	4,737	+0.00113	+0.00287
Human, 1 session	3,327	−0.00004	+0.00170
Human, 2 to 4 sessions	570	+0.00322	+0.00496
Human, 5 + sessions	840	+0.00435	+0.00609

Table 4. Paired ΔH_{RS} is the single-scale H_{RS} score for the Subject stream minus the score for the same-call PCS stream; its paired-null reference is zero. The Human session-count strata are mutually exclusive, and Human-all combines the three strata. The final column subtracts the observed Baseline paired mean of −0.00174 and is descriptive rather than a separate inferential test.

The observed paired means increased across the Human session-count strata, from -0.00004 in the one-session group to +0.00435 in the 5+ group. Because session count was confounded with recruitment source and participant characteristics, this pattern cannot be interpreted as evidence of learning or a dose-response effect.

The Human 5+ contrast with Baseline was +0.00609. The observed Baseline mean of -0.00174 contributed approximately 29% of that contrast, so the result depended partly on the particular Baseline realization. The 5+ threshold was also one of several retrospectively examined thresholds and produced the largest contrast. Its selection and calibration are examined in Section 3.6.4.2.

The AI Agent paired mean was +0.00094. Its unclustered comparison with Baseline produced $D = 0.0227,\ p = 0.3359$. No AI Agent versus Baseline distributional difference was detected at this resolution.

The descriptive variation across the Human session-count strata is shown in Figure 4.

Figure 4. Mean paired single-scale H_{RS} by condition and Human session-count group

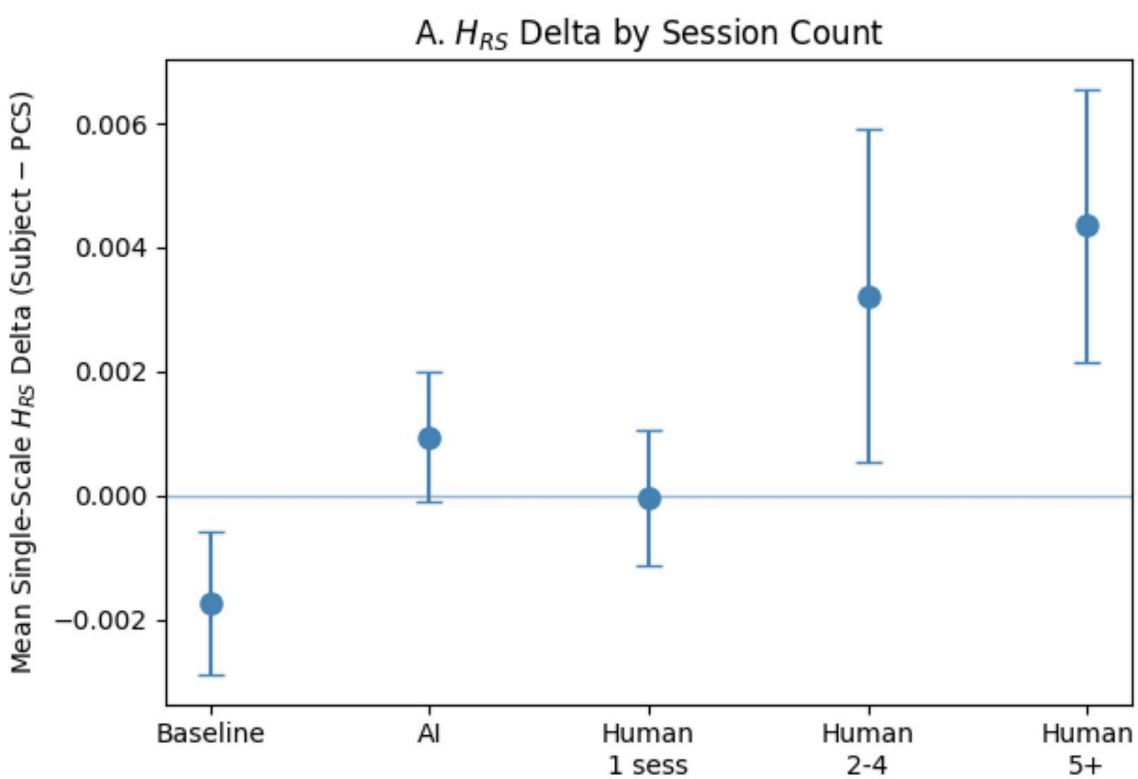

Figure 4. The Human 5+ subgroup had the largest observed mean among the Human strata. Error bars show block-level ±1 SE and are descriptive rather than participant-level uncertainty.

3.6.3 Shuffle Sensitivity to Within-Block Ordering

The H_{RS} score is calculated from the cumulative-deviation path of a sequence, so changing bit order can change the score even when the number of zeros and ones remains fixed. To test this property, the same positional permutation was applied to both streams within each paired block. This preserved bit composition and pairing while altering within-stream order.

The analysis used window sizes of 1, 3, 5, 10, 25, 50, and 150 bits. At $W = 1$, the ordering was unchanged. At $W = 150$, each complete 150-bit stream was fully shuffled. The Human 5+ versus Baseline distributional separation changed across these conditions and was smallest after full-block shuffling.

This shows that the observed H_{RS} separation was sensitive to within-block ordering rather than bit composition alone. The full window-specific results are reported in Supplementary Section S3.

3.6.4 Joint Interpretation of Direct and Paired Results

Direct and Paired-differential analyses use the same underlying stream scores but provide different views of a condition contrast. Their combination can show whether an observed difference is concentrated in Subject, shared with PCS, or formed by movement in both streams. Interpretation therefore requires examination of the separate Subject and PCS components, not only the statistical result of either comparison.

The general Direct and Paired-differential comparisons are defined in Section 2.6.1. Here, that framework is applied to the single-scale H_{RS} score as the pilot worked example. The paired quantity is

$$\Delta H_{RS} = H_{RS,Subject} - H_{RS,PCS}.$$

Table 5 summarizes how combinations of Direct and Paired-differential results can be interpreted.

Table 5. Interpretive framework for direct-stream and paired-difference results

Direct-stream result	**Paired-delta result**	**Reading**
Difference detected	Difference detected	The Subject stream differs between conditions, and the within-call Subject minus PCS relationship also differs. Interpretation depends on the direction and size of both results.
Difference detected	No difference detected	The Subject stream differs between conditions, but no difference in the within-call relationship was detected. The effect may be expressed similarly in both streams, or the paired analysis may have lower sensitivity.
No difference detected	Difference detected	The within-call Subject minus PCS relationship differs between conditions even though no direct Subject-stream difference was detected. This can occur if PCS differs between conditions or if pairing removes shared variation.

No difference detected	No difference detected	Neither analysis detected a difference at the available resolution.

Relational analysis is not included in Table 5 because it examines joint stream behavior rather than the level of either stream or their within-call difference. The following subsections apply the Direct and Paired-differential framework to the exploratory Human 5+ result.

3.6.4.1 Application to the Human 5+ Comparison

For the 5+ session subgroup, the unclustered direct comparison of Subject-stream H_{RS} scores between Human 5+ and Baseline ($H_{RS,Subject,Human}$ vs $H_{RS,Subject,Baseline}$) produced KS $D = 0.0575$, $p = 0.0243$. The corresponding comparison of the paired Subject-minus-PCS scores produced KS $D = 0.0541$, $p = 0.0398$. Both therefore showed nominal distributional differences in the same direction.

Because Direct and Paired-differential comparisons were both identified as analyses of interest, their joint interpretation also carries the multiplicity associated with testing two overlapping hypotheses. A two-dimensional Kolmogorov-Smirnov comparison of the joint Subject and PCS values produced $D = 0.0681$, $p = 0.0480$. This analysis is not independent confirmation of the Direct or Paired results.

Because the Human 5+ subgroup contained only three participants, these block-level tests do not provide reliable participant-level inference. Their value here is descriptive. They show that the observed separation was visible in both the Subject-stream and paired views of the same data.

The separate Subject and PCS contributions are examined in Section 3.6.4.2. The implications of post-generation stream assignment for interpreting a label-aligned result are discussed in Section 5.9.

3.6.4.2 Bidirectional Stream Pattern and Subgroup Selection

The hierarchy-preserving Human 5+–Baseline contrast in the mean paired H_{RS} was +0.006090, with a descriptive 95% bootstrap interval of $[+\ 0.000776,\ +\ 0.016003]$. The separate stream estimates showed how this contrast was formed. Subject was +0.00315 above Baseline, while PCS was -0.00294 below Baseline. The two components were therefore similar in magnitude and opposite in direction.

To understand why the delta was large in this subgroup, the individual stream distributions were examined separately. For the 5+ subgroup, the Subject stream (Human 5+) trends above baseline ($t = \ +\ 1.770$, $p =\ 0.077$) while the PCS (Human 5+) trends below baseline ($t = \ -\ 1.668$, $p =\ 0.096$). This configuration is inconsistent with the simplest common-mode pattern in which both streams shift equally and cancel under subtraction. It does not, however, establish differential stream behavior. When a subgroup is selected for having a large Subject-minus-PCS difference, weakly correlated streams with similar variability can naturally produce one component above and the other below the reference. The observed Subject and PCS split is compatible with that selection process.

The 5+ subgroup produced the largest result among the retrospectively examined 2+, 3+, and 5+ thresholds. A composition-conditional order-randomization analysis therefore evaluated both the fixed 5+ contrast and selection of the largest positive contrast across those three thresholds. The one-sided tail area was .00695 for the fixed 5+ comparison and .00800 after accounting for selection among the three thresholds. This calibration addresses selection within that threshold family only. The broader exploratory-search limitation is discussed in Section 5.3.

The result therefore remains compatible with both differential stream behavior and retrospective selection for a large paired difference. The implications of post-generation Subject and PCS assignment for interpreting a label-aligned pattern are discussed in Section 5.9.

3.6.5 Run-structure Check

Run structure was examined to characterize what the single-scale H_{RS} score measures in the observed 150-bit sequences. Blocks in the lower tail of the Baseline Subject-stream H_{RS} distribution generally had more runs and shorter maximum runs, whereas blocks in the upper tail had fewer runs and longer maximum runs. This pattern is consistent with the cumulative-excursion construction of the score and should be interpreted as characterization rather than independent validation.

A separate hierarchy-preserving paired analysis compared Human 5+ with Baseline using run count, maximum run length, and run-length entropy. The contrasts were -0.058 runs for run count, +0.048 bits for maximum run length, and +0.00089 for run-length entropy. All 95% descriptive intervals included zero. These simpler run-based measures therefore did not reproduce the Human 5+ paired H_{RS} contrast at the available resolution.

Full results are reported in Supplementary Section S3.

3.6.6 Bit Composition

Order-1 Shannon entropy was used as a measure of bit balance. The paired Human 5+ Subject-minus-PCS entropy difference was −0.00036 $(95\%\ descriptive\ interval:\ -\ 0.00111\ to\ +\ 0.00070)$. Relative to Baseline, the Human 5+ paired contrast was −0.00044 $(-\ 0.00127\ to\ +\ 0.00068)$. Both intervals included zero. No detected bit-composition imbalance therefore accompanied the Human 5+ paired H_{RS} pattern.

3.6.7 Variance Comparison

The block-level variance of the paired H_{RS} score was similar in Human 5+ and Baseline. Variance was 0.004091 in Human 5+ and 0.004116 in Baseline, producing a variance ratio of 0.994 with a 95% descriptive interval of 0.865 to 1.172. The interval included 1.0, providing no clear evidence that increased paired-score variability accompanied the Human 5+ mean difference.

3.6.8 Interpretation

Taken together, the exploratory checks did not identify a simple explanation for the Human 5+ pattern. Several possibilities remain compatible with the data.

The opposed Subject and PCS directions could reflect genuine differential stream behavior, but the same configuration can arise from retrospective selection for a large Subject-minus-PCS difference. The threshold-selection calibration reduces that concern within the documented 2+, 3+, and 5+ comparisons but does not account for the broader exploratory search. The within-block label-swap analysis showed that the observed alignment of Subject and PCS labels with the paired scores was stronger than expected under label exchange, but it evaluates the same paired data and does not identify the source of that alignment. Full threshold and label-swap results are reported in Supplementary Section S4.

The shuffle analysis provides a different piece of information. The Human 5+ versus Baseline separation changed when within-block ordering was disrupted while bit composition was preserved, with the smallest separation after full-block shuffling. This indicates that the measured pattern depends on sequence ordering rather than simple bit balance alone, but it does not establish why that ordering difference arose. Full shuffle results are reported in Supplementary Section S3.1.

Interpretation is complicated by several competing explanations. Retrospective subgroup selection can produce the observed opposed Subject and PCS directions, while the label-swap result shows that the recorded stream labels carried structure in the paired data without identifying its source. The shuffle analysis shows that the separation depends on within-block ordering rather than bit composition alone. Finite-sample behavior of the H_{RS} score could also contribute, although the simulations in Section 5.10 did not show estimator asymmetry large enough to explain the full contrast within the models examined. A DFA cross-check showed the same direction with a smaller standardized difference, while other ordering measures did not clearly reproduce the contrast. Full shuffle and metric-cross-check results are reported in Supplementary Sections S3.1 and S3.4, and threshold and label-swap analyses are reported in Supplementary Section S4.

The available evidence therefore does not distinguish among retrospective selection, the particular Baseline realization, metric-related distortion, a reproducible ordering difference, or some combination of these factors. The pilot identifies a candidate pattern and shows how PDP can decompose and interrogate it, but it does not establish its cause.

3.7 Additional Robustness Checks

3.7.1 Sensitivity to Position, Weighting, Recurrence, and Session Order

Several additional checks examined whether the Human 5+ H_{RS} pattern depended on within-session position, contributor weighting, individual sessions, or session order. The continuous position-slope analyses were unresolved in every condition. The Human 5+ minus Baseline contrast remained positive under block weighting and equal-contributor/equal-session weighting, and its direction did not depend on the principal investigator or any single session. First-versus-later-session comparisons showed mixed contributor-specific trajectories. Full results are reported in Supplementary Section S1.

A post hoc split-half diagnostic showed that the negative Baseline realization was concentrated more strongly in the first half of sessions, which contributed to the magnitude of the Human 5+ minus Baseline contrast. However, the direct comparison of the condition contrast between session halves was unresolved, providing no clear evidence of a systematic position effect. Full split-half results are reported in Supplementary Section S1.

3.7.2 Relational Capability: Local and Lagged Coupling

Beyond the pooled relational analyses in Section 3.2, two exploratory relational patterns emerged from different relational estimands and temporal scales.

In the Human condition, exploratory analyses suggested lower variability in local Subject–PCS coupling across multiple window sizes, while average coupling remained near zero. This pattern concerns the stability of the relationship between the two streams rather than its mean strength. More recent exploratory work has treated this as a pattern across the window-size curve rather than as a finding tied to one particular window. Related session-level reproducibility patterns were examined separately. Full exploratory details are reported in Supplementary Section S2.1.

Separately, in the full Human condition, lagged Subject–PCS cross-correlation analysis identified lower between-session variability at lag +2. This result survived correction within the 14-comparison lag family, with a conservative adjusted $p = .01560$. No individual lag had been predicted in advance, and the result remains exploratory within the broader pilot search. Full lagged analyses are reported in Supplementary Section S2.2.

Both findings concern the stability of the relationship between Subject and PCS rather than a mean shift in either stream. They illustrate the distinct information available from Relational analysis, but they should not be treated as mutually confirming evidence of a single phenomenon because they involve different temporal scales and estimands.

3.8 Preregistered Synthetic Diagnostic-Recovery Validation

A separate held-out synthetic validation evaluated whether the frozen PDP diagnostic rules recovered deliberately constructed patterns with known generating properties. Diagnostic classifications were produced without access to the generating-condition key and were compared with that key only after the prediction file had been frozen.

The marginal-only condition was correctly identified in 300 of 300 held-out datasets, and the zero-lag relational-only condition was correctly identified in 300 of 300. Temporal localization was correct in all 1,500 of 1,500 datasets containing an injected cross-stream relationship. Each registered lag, −2, −1, 0, +1, and +2, was correctly localized in 300 of 300 datasets. The coupling-instability condition was correctly identified in 100 of 100 held-out datasets using the frozen 10-block criterion.

The 100 null-control datasets produced no marginal, zero-lag relational, or coupling-instability false-positive flags. No claim was made about no-lag classification because the preregistration had not specified a decision rule for absence of lagged dependence.

Across all 2,300 held-out datasets, every applicable preregistered diagnostic criterion was met. These results show that the frozen implementation recovered the targeted statistical properties under the synthetic conditions tested. They do not establish diagnostic performance for independently arising or unknown disturbances in empirical data.

4. Discussion

4.1 Architectural Contribution

The Paired-Delta Protocol extends the conventional single-stream design. Each acquisition retains a Subject stream and a same-call Paired Control Stream, allowing the same data-generating event to be viewed in three complementary ways. Direct analysis preserves the conventional comparison of a stream across conditions. Paired-differential analysis adds a within-call contrast that shows how Subject and PCS contribute to an observed condition difference. Relational analysis adds a joint-information branch that examines how the two streams vary together. The architecture therefore retains the analyses available in a single-stream design while adding paired and relational analyses that require two streams

Employing a same-call PCS provides information that is unavailable from a conventional single-stream design. If the Subject stream differs between Human and Baseline conditions, the PCS shows whether a comparable change also occurred elsewhere in the same acquisition and how the two streams contributed to the observed contrast. A paired contrast can therefore distinguish several configurations that would look similar if only the Subject stream were retained. The condition difference may be concentrated in Subject, concentrated in PCS, shared across both streams, or produced by movement in opposite directions. This does not identify the cause of the difference, but it provides a more complete description of where the observed contrast resides.

The paired-differential branch also provides conditional common-mode control. Contributions expressed similarly in the selected Subject and PCS statistics can be attenuated by subtraction, whereas unequal contributions remain. This is useful for diagnosing acquisition-time variation shared by the two measurements, but it is not a universal artifact-removal procedure. A genuine condition-related change could also affect both streams and therefore be reduced by subtraction. For that reason, Direct and Paired-differential analyses answer related but non-identical questions and are most informative when their component streams are examined together.

The Relational branch adds a structurally different kind of information. Two streams can each show unremarkable marginal behavior while differing in their correlation, covariance, temporal alignment, stability, or other joint properties. A condition-related difference could therefore appear in how Subject and PCS co-vary even when neither stream shows a clear mean-level shift. Relational statistics do not require that such a difference be expressed as a directional change in either stream. Because these analyses use overlapping observations from the same acquisitions, however, relational findings should not be treated as statistically independent confirmation of Direct or Paired-differential findings.

The architecture is also metric-agnostic. Any appropriately calibrated stream-level statistic can be evaluated directly and, where meaningful, as a paired difference, while statistics defined jointly from Subject and PCS can support Relational analysis. The single-scale H_{RS} score served as the pilot's worked example rather than as a defining feature of PDP. The architectural contribution comes from the additional observational structure created by retaining and analyzing two same-call streams.

4.2 What the Pilot Data Suggest

The pilot did not provide a confirmatory test of observer-linked deviations from chance. No clear paired mean hit-rate difference was detected, and the principal ordering-sensitive result emerged retrospectively in the Human 5+ subgroup. The resulting H_{RS} pattern therefore remains a candidate finding rather than evidence of a reproducible population effect.

What the paired architecture added was diagnostic resolution. The Human 5+ contrast involved both streams. Subject was above Baseline while PCS moved in the opposite direction, and the realized Baseline difference also contributed materially to the overall contrast. The same observed paired result was therefore compatible with several explanations, including differential stream behavior, retrospective selection, the particular Baseline realization, sampling variation, and metric-related effects. The pilot analyses did not distinguish among them.

The exploratory relational analyses suggested an additional possibility. Condition-related structure may appear not only as a shift in an individual stream or in the Subject-minus-PCS difference, but also in the stability or temporal alignment of the relationship between the two streams. These relational findings remain exploratory, but they identify joint-stream properties that would not be observable in a conventional single-stream design.

Taken together, the pilot does not establish a specific observer-related effect. Its empirical contribution is narrower. It identified candidate ordering and relational patterns while showing how the paired architecture can expose alternative explanations and component structure that would remain hidden if only the Subject stream were retained.

4.3 Analytical Value of the Three Analysis Families

PDP supports three analysis families organized into two broader analytical branches. The stream-level branch includes Direct and Paired-differential analysis. Direct analysis evaluates Subject and PCS separately across conditions, while Paired-differential analysis evaluates their within-call difference. Taken together, these quantities show whether an observed condition difference is concentrated in Subject, concentrated in PCS, shared across both streams, or formed by movement in opposite directions. The joint-information branch consists of Relational analysis, in which the statistical object is a property of the pair itself rather than either stream or their difference.

This distinction matters because ordinary stream-level behavior does not imply ordinary joint behavior. Two streams can each show similar marginal means, variances, or other stream-level properties across conditions while differing in how they covary, how stable that association is, or how their fluctuations align over time. Correlation, covariance, and mutual information are symmetric under exchange of the stream labels, whereas directional measures such as signed lagged cross-correlation retain information about which stream occupies each role. Relational analysis therefore asks a different kind of question from either Direct or Paired-differential analysis.

The idea that an observer-related difference might appear in relationships among variables rather than as a stable shift in a single outcome has precedent in the work of von Lucadou, Grote, and Walach (von Lucadou et al., 2007; Grote, 2017;

Walach et al., 2022). The Correlation Matrix Method examines whether putative effects may be distributed across patterns of relationships rather than concentrated in a single outcome variable. Grote (2017) extended this relational approach to associations between psychological variables and binary random events, while Walach et al. (2022) similarly considered whether putative effects may be distributed across the experimental system rather than localized to a single measured outcome.

PDP draws on this relational motivation but differs substantially in structure and scope. It does not implement the Correlation Matrix Method and does not test the Model of Pragmatic Information. Rather than constructing a matrix of relationships among many psychological and physical variables, PDP asks a much more narrow question about whether two physical streams produced within the same QRNG acquisition differ across conditions in their joint behavior. The relevant quantities can include association, temporal alignment, local stability, and features of the joint distribution.

Von Lucadou's Model of Pragmatic Information further specifies a boundary condition on any such distributed effect summarized as the model's nonlocality axiom (NT axiom). It states that a statistically reliable signal between an emitter and a receiver, once established, should decline under continued observation or replication, regardless of whether that signal is expressed as a mean-level effect or as a relational one. The paired architecture's relational branch is not structurally exempt from this prediction. A stable correlation between Subject and PCS that reliably distinguished Human from Baseline conditions would satisfy the same definition of a decodable signal as any single-stream effect, and would therefore be subject to the same predicted decline.

The same-call structure is important to this distinction. Conventional control observations collected at another time can characterize system behavior, but they do not share the immediate acquisition history of the experimental observation they are intended to contextualize. Subject and PCS do. This makes it possible to ask whether the relationship between two parts of the same acquisition differs when the experimental condition is present. This same-call structure makes it possible to test whether condition-related information appears in the relationship between Subject and PCS rather than in either stream alone. That possibility is consistent with earlier proposals that putative effects may be distributed across relationships within an experimental system rather than concentrated in a single outcome (von Lucadou et al., 2007; Grote, 2017; Walach et al., 2022). PDP does not assume those models are correct and does not test them directly.

The pilot illustrates the two branches differently. The stream-level branch decomposed the candidate Human 5+ H_{RS} contrast into its Subject, PCS, and Baseline components. The Relational branch identified exploratory differences in the variability and temporal alignment of Subject–PCS coupling. Average coupling itself remained near zero, emphasizing that a relational difference need not take the form of a shift in mean association. These exploratory findings do not establish a common mechanism or a relational observer effect. Their value is that they identify joint-stream questions that can be specified prospectively in later work.

Relational analysis also extends beyond the particular statistics examined in the pilot. The local-window analyses ask whether the stability of Subject–PCS association changes across temporal scales, while lagged analyses ask whether the alignment between the streams changes across temporal offsets. Other joint quantities could characterize persistence, changes in sign, nonlinear dependence, or the coincidence of extreme values. Relational analysis is therefore not a single test but a broader class of questions made possible by retaining two same-call streams.

4.4 Planned Pre-Registered Replication and Permeability Framework

The pilot therefore leaves two kinds of questions for prospective study. One concerns the defined candidate ordering pattern, and the other concerns a broader set of exploratory relational questions. A follow-up study is planned and will be preregistered before data collection begins. The primary confirmatory test will be a direct, two-tailed comparison of Human and Baseline Subject-stream single-scale H_{RS} scores, without assuming in advance whether any difference will reflect greater persistence or antipersistence. Paired-differential and PCS comparisons will be included as mandatory diagnostics to show how any observed Human–Baseline difference is expressed across the two streams.

The follow-up will also continue the relational questions raised by the pilot. Exploratory analyses suggested that local Subject–PCS correlations may vary less within Human sessions than within Baseline sessions, even though their average correlation remained near zero. The planned analysis will therefore examine the window-to-window variability of local Subject–PCS correlations across several prespecified window sizes. Temporal-lag relationships will be explored separately across a prespecified range of lags. These relational analyses remain exploratory. The pilot's more detailed relational results and their limitations are discussed in Section 4.6.

Several features of the follow-up are intended to address limitations that became clear in Study 1. Pilot sessions contained only 30 blocks, and most participants completed a single session, leaving highly unequal contributions across participants and limited information about participant-level variability. The follow-up will use longer, fixed-length sessions and repeated participation so that each participant contributes a more comparable body of data. Population-level inference will explicitly preserve the hierarchy of blocks within sessions and sessions within participants rather than treating the much larger number of blocks as independent observations. These changes are intended to improve both statistical resolution and the basis for participant-level inference.

The Baseline design will also be strengthened. In the pilot, Baseline was collected separately from much of the Human data, and provider identity was not completely retained at the block level. In the follow-up, Baseline collection will run contemporaneously with Human participation, and QRNG provider exposure, timing, failures, and other acquisition information will be recorded prospectively. The same provider configuration will be used across Human and Baseline as far as practicable, reducing the provider and collection-period confounding that complicated interpretation of the pilot.

The study will also examine individual differences within a provisional permeability framework, defined as reduced filtering at the boundary between the person and available information. Candidate predictors include psi belief, self-assessed ability, anomalous experience, meditation practice, the MAIA-2, and the BQ-18, with the HSP-12 and Revised Transliminality Scale added if permission is obtained. Because no validated permeability measure currently exists, these predictors will be analyzed separately and exploratorily.

The follow-up will use longer fixed-length sessions and repeated participation, with analyses accounting for blocks within sessions and sessions within participants. Baseline data will also be collected contemporaneously with Human data. The design, analysis plan, and confirmatory procedures will be reviewed and fixed prospectively before outcome data are examined.

4.5 The Single-Scale H_{RS} Ordering Score as a Metric

The single-scale H_{RS} score served as an exploratory ordering-sensitive screening statistic (Section 2.6.3). Run-structure analysis helped characterize what the score detects. Blocks with lower H_{RS} values generally contained more runs and shorter maximum runs, whereas blocks with higher values had fewer runs and longer maximum runs. Longer same-value runs tend to produce the larger cumulative excursions measured by the single-scale H_{RS} score, so this association was expected rather than an independent validation of it. Paired run-count, maximum-run-length, and run-length-entropy contrasts did not reproduce the Human 5+–Baseline H_{RS} contrast at the available resolution (Section 3.6.5). H_{RS} therefore appears to summarize aspects of ordering that were not captured fully by any one of these simpler measures.

The score was also sensitive to the simulated persistence structure examined in the candidate-scale injection analysis. When an ordering difference was deliberately introduced, the resulting paired H_{RS} contrast was recovered at a scale comparable to the pilot pattern. This supports its use as an ordering-sensitive measure for the particular structures tested, while not establishing sensitivity to every form of temporal organization. Within PDP, the same score can be examined directly in Subject or PCS, as a Subject-minus-PCS difference, or relationally by studying how the two streams' H_{RS} values vary together.

The interpretation of H_{RS} depends on sequence length and on the form of ordering present in the data. Because the pilot calculated the score at a single 150-bit scale, it is not interpreted as a conventional Hurst exponent or as evidence of long-range dependence. Its finite-sample behavior and the limits of the calibration simulations are discussed in Section 5.10.

More generally, when temporal ordering is part of a PDP research question, the ordering measure should be selected prospectively and calibrated for the sequence length and structure being studied. Hit rate and H_{RS} illustrate different kinds of information available from the same streams. Hit rate summarizes target-aligned bit composition, while H_{RS} is sensitive to aspects of within-block ordering. Neither measure by itself identifies the mechanism responsible for an observed condition difference.

4.6 Bit Composition, Temporal Ordering, and Relational Structure

The pilot illustrates why bit composition, temporal ordering, and relational structure should be treated as different properties of the same QRNG data. A sequence can contain an ordinary balance of zeros and ones while arranging those bits in an unusual order, and two streams can each appear ordinary on their own while differing in how their local behavior changes together. PDP makes all three levels available for comparison.

At the composition level, the pilot found no clear condition differences in overall bit balance or short-pattern frequencies (Section 3.6.6). The single-scale H_{RS} result pointed instead toward ordering. Because H_{RS} depends on the path traced by the sequence rather than only on how many zeros and ones it contains, the same bit composition can produce different scores when its order changes. The shuffle analyses supported this distinction, with the exploratory Human 5+ separation becoming smaller as within-block order was disrupted. The candidate H_{RS} pattern therefore appears more closely tied to temporal organization than to a simple excess of one bit value.

The relational analyses opened a third level of description. Here the question was not whether Subject or PCS had a higher score, but whether the local relationship between them behaved differently across conditions. In the Human 5+ subgroup, local Subject–PCS correlations were less variable than Baseline at window sizes 5 and 10. The pilot's 30-block sessions limited this analysis to those shorter windows because larger windows left too few observations within each session to estimate window-to-window variability reliably. In the full Human sample, a separate lagged analysis found lower between-session variability at lag +2, surviving correction within the tested lag family. That lag-specific pattern did not reproduce in the separate dataset.

These relational results describe different temporal features and should remain separate rather than being collapsed into a single effect. What they have in common is more modest but methodologically useful. They show that condition-related structure can be sought not only in the level or ordering of either stream, but also in the stability and temporal organization of the relationship between them. The pilot therefore moved the analysis from asking only *how many* target-aligned bits occurred, to how those bits were ordered, and finally to how two same-call streams behaved in relation to one another. The resulting relational questions remain exploratory and provide specific candidates for prospective study rather than established effect.

4.7 Architecture Parameters

The design choices reported here reflect one configuration among several the architecture supports. Four parameter classes are worth distinguishing because they are design settings rather than fixed features of the Paired-Delta Protocol.

Timing and partitioning. The timing of stream-label assignment relative to bit generation affects the mechanistic interpretation of a stream-specific difference. In the pilot, labels were determined from an assignment bit generated in the same call and applied after generation. A design that assigns Subject and PCS identities before content generation would remove this particular timing ambiguity. The implications of assignment timing are discussed in Section 5.9.

Stream partitioning is a separate design choice. The pilot divided each 300-bit sequence into two contiguous 150-bit halves, so Subject and PCS occupied different portions of the acquisition interval. This makes the paired comparison potentially sensitive to systematic changes across the duration of a call. In a retrospective reconstruction of the Baseline data, distributing both streams across the acquisition interval through locally balanced interleaving substantially attenuated the session-position pattern observed with the original contiguous split. Because the alternative partition was evaluated after that pattern was identified, this result does not establish that interleaving is generally superior or that the original pattern was artifactual. It does show that partitioning strategy can affect sensitivity to within-call temporal structure and should be treated as an explicit design parameter. Interleaving is not part of the current Study 2 replication design but remains a candidate configuration for future prospective testing.

Measurement. The single-scale H_{RS} score was one choice of ordering metric, matched to the 150-bit block length used here (Section 4.5). The architecture does not privilege this metric, and alternative measures may differ in their sampling behavior, sensitivity, precision, and bias. Stream length is likewise a measurement choice because it affects the behavior and resolution of the statistic being calculated. The 150-bit length used in the pilot reflected feasibility constraints rather than evidence that this length was optimal. Longer streams may improve the finite-sample behavior of H_{RS} and increase the resolution of some relational measures, although this has not yet been established using genuine longer sequences. The empirical physical-half comparison in Section 3.2.2 therefore applies specifically to the two contiguous 150-bit halves examined here and should not be generalized to other sequence lengths or partitioning rules.

Any stream-level metric can, in principle, be calculated separately for Subject and PCS and then examined through Direct or Paired-differential analysis. Metrics defined jointly from the two streams can support Relational analysis. Each metric and stream configuration therefore requires its own calibration, including evaluation of finite-sample behavior, sensitivity, precision, and appropriate null distributions. The finite-sample behavior of the pilot's single-scale H_{RS} implementation is discussed in Section 5.10.

Acquisition structure. The number of blocks per session, number of sessions contributed by each participant, and audit frequency can also be varied without changing the paired architecture. These choices affect statistical resolution and the amount of information available at the block, session, and participant levels. The pilot's short sessions and highly unequal participant contributions illustrate why these parameters should be chosen with the intended analysis and unit of inference in mind. In particular, relational analyses based on local windows require enough blocks within a session to estimate how the relationship changes across windows. The 30-block pilot sessions restricted the local correlation-variability analysis to relatively short windows, whereas longer sessions can support examination across a broader range of temporal scales.

Task. Several features of task presentation can be changed without altering the paired architecture, including whether feedback is provided and how the target, feedback, or reward is presented. More substantial alternatives could change what information is available during acquisition, such as displaying the developing bit sequence in a ring or matrix rather than presenting only aggregate feedback. The task could also be framed around objectives other than intentional influence. These choices may affect engagement, learning, strategy, and the interpretation of any observed difference.

5. Limitations

5.1 Short protocol duration

Sessions were designed to contain 30 blocks, with two 150-bit streams generated per block. This brief format was constrained by available funding for participant compensation and QRNG API calls. It prioritized accessibility but limited statistical resolution and increased the sampling variability of block-level metrics relative to protocols using longer streams or more blocks per session.

5.2 Post-hoc subgroup analysis

The Human 5+ subgroup was initially examined because participants recruited from the Reddit remote-viewing community had been encouraged to complete five to ten sessions, making session count an available but imperfect proxy for that recruitment group. The broader 2+ and 3+ thresholds were subsequently examined to explore whether the pattern might instead reflect repeated participation or learning. Human 5+ produced the strongest result among the three nested thresholds and is therefore emphasized here, with multiplicity across the threshold analyses addressed as described in Section 3.6.4.2. Because recruitment source, repeated participation, and self-selection are confounded, and Human 5+ includes only three participants, the subgroup finding remains exploratory.

5.3 Scope of the Exploratory-Search Calibration

The composition-conditional order-randomization analysis accounted for selection of the largest positive contrast across the three documented 2+, 3+, and 5+ session thresholds. The limitation is that this was a bounded calibration rather than an adjustment for the complete exploratory program. It did not include the wider historical search across metrics, populations, label-swap analyses, disjoint session-count groups, or relational measures. The calibration therefore strengthens interpretation of the result within the documented threshold family but does not convert the Human 5+ finding into a confirmatory result. The relational analyses were also exploratory, with their search scope and multiplicity limitations described in Section 4.6.

5.4 Recruitment confound

Higher session-count participants were recruited from a different source than most single-session participants. Session count is therefore confounded with recruitment channel and participant characteristics. The present data cannot distinguish between exposure effects, trait differences, or self-selection.

5.5 Subgroup Size and Experimenter Participation

The 5+ session subgroup consists of three participants (~840 blocks), which limits what block-level analyses can establish about generalizability across observers. Clustered bootstrap resampling (Section 3.7.1) preserves the positive direction of the effect but widens confidence intervals substantially relative to block-level resampling, reflecting this small contributor count. One member of the subgroup was also involved in study administration. Although the paired architecture limits mechanical influence on outcomes, inclusion of a non-blinded experimenter limits independence of the subgroup. Excluding this participant (Section 3.7.1) increases the effect size, because the principal investigator's blocks show the smallest individual paired-delta among the three.

5.6 Statistical resolution limits

As reported in Section 3.5, the Human-all versus Baseline comparison had approximately 80% power to detect a paired hit-rate contrast of 0.372 percentage points. This is substantially larger than the absolute hit-rate deviations of approximately 0.02 percentage points reported in the classical RNG-intention literature (e.g., Jahn et al., 1997; Bösch et al., 2006). This comparison provides only approximate scale context because the historical absolute deviation from chance and the present Human-minus-Baseline paired-delta contrast are different estimands.

The pilot was therefore not powered to reliably detect effects on that historical hit-rate scale. Its unresolved population-level results should not be interpreted as evidence that smaller effects are absent. More generally, underpowered individual studies have a low probability of detecting small effects even when such effects are present (Maxwell, Lau, & Howard, 2015).

Statistical resolution was also limited for the paired H_{RS}-score analysis. Under the session-cluster bootstrap location-shift model, the Human-all versus Baseline comparison had approximately 80% power to detect a contrast of 0.00405 H_{RS}-score units. The observed contrast was smaller (+0.00287), and its interpretation was borderline and analysis-method-sensitive: the hierarchy-preserving bootstrap interval only narrowly included zero, whereas the block-level Welch test did not reach the conventional significance threshold (Section 3.6.1). The study therefore had limited power to reliably detect H_{RS}-score contrasts of the observed magnitude.

Greater precision requires larger samples, pooled evidence, or repeated independent studies, motivating the higher-powered replication planned for Section 4.4, to be preregistered before data collection.

5.7 AI control implementation

The AI condition combined GPT-4o-mini responses with scripted browser automation. Although completion of the model's response gated each QRNG call, the agent could not deliberately pause, select an initiation time, or otherwise exercise the open-ended self-pacing available to human participants. It also had no persistent memory or model updating across blocks or sessions. The condition should therefore be interpreted as a constrained, API-driven procedural control rather than a fully adaptive agent with independent control over task timing or strategy.

5.8 QRNG Provider Heterogeneity and Execution Environment

QRNG calls were served through more than one API provider during the study period, but provider identity was not successfully recorded at the block level. Provider exposure therefore could not be balanced or included as an analytic factor. Paired subtraction removes provider-related contributions only when they are expressed equally in the two stream statistics; unequal, position-dependent, or condition-associated provider differences may remain. The three-subtest audit battery showed similar pass rates across conditions, but its limited scope and the partial confounding of provider, date, and collection regime prevent it from excluding all provider-level differences.

Collection structure presents a related limitation. Baseline data were collected in larger batches, whereas Human data were generated through many short sessions occurring at different times. Consequently, condition was not fully separable from collection cadence and timing. The available audit and session-structure analyses found no corresponding instability or explanation for the observed subgroup pattern, but they did not directly test every possible effect of batch size, time of day, or longer-term drift.

Human participants also accessed the experiment through web browsers on uncontrolled local devices and networks. Latency and device differences were not explicitly modeled. Because both streams came from the same API response, paired subtraction would remove any resulting contribution expressed equally in both stream statistics, but it cannot exclude unequal or interaction-dependent effects.

Future studies should record provider identity for every block, balance provider exposure across conditions, and collect Baseline data in session-length batches staggered to match the timing and cadence of Human sessions. Device, browser, latency, and relevant timing variables should also be logged when privacy and feasibility permit.

A dedicated follow-up validation study has since directly tested this gap using controlled, per-block-logged provider identity (Section 3.3.5). It found no significant provider-driven difference in hit rate or H_{RS}, with a resolution sufficient to rule out an effect near this study's own benchmark magnitudes, though not sufficient for full equivalence certification at the finest resolution tested. This does not retroactively recover provider identity for the pilot's own blocks, which remains the limitation described above, but it directly bounds how large an uncontrolled provider effect could plausibly have been.

5.9 Timing of stream assignment

The Subject and PCS labels were determined after generation of each 301-bit call. Bit 0 was generated in the same call as the two 150-bit halves and was subsequently read by the software to determine which physical half would be labeled Subject. Consequently, the halves had fixed physical positions during generation but did not yet have fixed Subject and PCS identities. Random assignment protects against a stable difference between the first and second halves systematically aligning with the Subject label, although it does not exclude chance imbalance, implementation error, or an artifact associated with the assignment bit itself.

This timing constrains the mechanistic interpretation of a stream-specific result. An observed difference aligned with the subsequently assigned Subject stream would require either an influence associated jointly with the assignment bit and the corresponding half, or some other process capable of coordinating the generated values with their later labels. By contrast, an influence acting uniformly on the complete call would not require stream identity to exist during generation, but it would ordinarily affect both subsequently labeled streams and cancel from the paired difference. This cancellation property holds for any paired-difference design regardless of when stream identity is assigned. A whole-call influence should therefore be examined through Direct analysis of the separate streams because it could be reduced or cancelled in the paired difference. The Human 5+ result illustrates why the separate streams matter. In that subgroup, Subject was above Baseline while PCS was below Baseline, so the paired difference was formed by movement in both streams. Looking only at the delta would not show how that contrast was assembled. More generally, opposite-direction changes in Subject and PCS can enlarge the paired difference, while shared changes can reduce it. Examining Subject and PCS separately is therefore necessary to understand what produced the observed contrast.

A holistic model in which the assignment bit and both physical halves emerge as components of a jointly constrained outcome offers one possible interpretation, but the present design cannot test that model or distinguish it from position-dependent, assignment-related, implementation-based, or chance explanations.

Assignment timing does not invalidate the label-swap sensitivity analysis reported in Supplementary Section S4. That test evaluates whether the observed alignment between the randomized labels and the paired stream scores is stronger than expected under label exchange. However, any genuine label-aligned result, whether identified by the original paired comparison or its label-swap equivalent, would inherit the same mechanistic timing question. Symmetric relational measures, such as same-window correlation or mutual information, do not depend on which stream is called Subject, whereas directional lagged relationships must be interpreted with care because exchanging the streams reverses the sign of the lag.

A design assigning the stream identities before generation would remove this particular ambiguity because the physical halves would already have Subject and PCS roles when their bits were produced. A direct comparison of pre-assigned and post-assigned conditions could therefore test whether assignment timing changes any observed stream-specific result.

5.10 Finite-Sample Behavior of the Single-Scale H_{RS} Score

The single-scale H_{RS} score is a transformed, finite-sample statistic derived from rescaled-range analysis and is used here as an ordering-sensitive measure. It is not a conventional multi-scale estimate of the Hurst exponent and should not be interpreted as a precise or unbiased estimate of long-range dependence. Its behavior can depend on sequence length, bit composition, and the process producing the ordering structure. Consequently, equal estimator behavior under Baseline data does not establish that bias would remain equal if Subject and PCS had different underlying ordering structures.

A bounded simulation evaluated the exact 150-bit score using rank-binarized Gaussian AR(1) and fractional-Gaussian-noise sequences, while reproducing the observed Human 5+ Subject and PCS composition margins. The measured H_{RS}-score difference followed the imposed ordering contrast. It was positive when the stronger structure was assigned to Subject and negative when that structure was assigned to PCS. Across the examined parameter pairs, no resolved forward-versus-reverse distortion was detected. All pointwise 95% intervals for the directional asymmetry included zero, and the largest interval endpoint was approximately 0.00152, compared with the observed Human 5+ paired difference of +0.00435. Within these simulated cases, differential estimator asymmetry therefore did not account for the full observed contrast.

This calibration is nevertheless limited to the two generating models, parameter values, 150-bit sequence length, and composition margins examined. The process underlying the QRNG sequences is unknown, so the simulation neither supplies a bias correction nor proves that differential estimator behavior is absent under other forms of ordering structure. The observed paired H_{RS} score difference should therefore be interpreted as an uncorrected, metric-specific contrast rather than as a bias-free estimate of a latent difference in long-range dependence. This limitation concerns the selected metric rather than the paired-delta architecture, which can accommodate other calibrated stream-level or relational statistics. A related precision limit applies to the artifact-injection tests themselves. Both persistence dose 0.35 conditions remained indeterminate on the H_{RS} statistic under the preregistered δ=0.002 bound. The independent-position result was known before registration and disclosed under Prior Knowledge; the synchronized result was prospective. In both conditions, interval width was primarily limited by the 103 real Baseline sessions rather than by simulated realization count. Resolving this requires additional real Baseline sessions, not additional simulated realizations. The implications of these finite-sample properties for metric selection and stream length are discussed in Sections 4.5 and 4.7.

5.11 Contiguous Stream Partitioning

The implemented protocol divided each 300-bit experimental sequence into two contiguous 150-bit halves, so the paired streams occupied different portions of the acquisition interval. Any position-related change within a call could therefore contribute unequally to the two stream statistics and would not necessarily be removed by their paired difference.

In a retrospective Baseline diagnostic, reconstructing the streams through locally balanced interleaving substantially attenuated the early-versus-late pattern observed under the contiguous split. The result was similar across 500 pseudorandom reconstructions and under a deterministic odd-even allocation. However, the alternative partition was examined after the pattern had been identified and used the same data in which it was first observed. The diagnostic therefore does not establish that the original pattern resulted from contiguous partitioning or that interleaving is generally superior. The absence of the same resolved early-versus-late pattern in the Human and AI conditions also remains unexplained. Stream partitioning should therefore be treated as a limitation of the pilot. Interleaved or otherwise time-balanced allocations should be evaluated prospectively in future implementations rather than inferred to be superior from the retrospective reconstruction alone. Interleaving would also more closely match the differential-measurement analogy introduced in Section 4.1. A conventional beam splitter divides one beam into two paths that traverse the same interval simultaneously, whereas the pilot's contiguous partition assigns Subject and PCS to sequential rather than simultaneous portions of the acquisition.

6. Conclusion and Recommendations

The Paired-Delta Protocol was developed for experiments in which any condition-related difference is expected to be small relative to the ordinary variation of the random system. Under those conditions, a result is only as useful as the information available to interpret it. Detecting a difference is only part of the problem. The harder question is how much can be learned about where that difference appears and what alternative explanations remain plausible.

A conventional single-stream experiment gives the researcher one measurement from each acquisition. If that stream differs between conditions, the acquisition itself offers little help in showing where the difference sits or how the rest of the call behaved. PDP retains both the Subject stream and a same-call PCS. This allows the Subject result to be examined directly against Baseline, decomposed through the Subject-minus-PCS contrast, and studied relationally through the behavior of the two streams together. Each acquisition therefore carries more information for interpreting any condition-related difference.

The pilot provided a concrete example of why this matters. In the retrospectively identified Human 5+ subgroup, the Subject-stream single-scale H_{RS} score was above Baseline, while PCS was displaced in the opposite direction by a similar magnitude. A conventional design retaining only the Subject stream would have shown the Subject difference without revealing how the paired contrast was assembled. PDP showed the contribution of both streams, giving a fuller view of the observed pattern and more ways to interrogate it. The result remains exploratory because it arose in three retrospectively selected contributors and is subject to the limitations described in Section 5, but as a worked example it demonstrates the diagnostic information gained from the same-call companion stream.

PDP also opens a second class of questions through Relational analysis. Instead of asking only whether Subject or PCS differs across conditions, Relational analysis asks whether the relationship between the two streams changes. This can include their stability, temporal alignment, covariance, or other forms of joint behavior, even when the individual stream averages look ordinary. The pilot's windowed-correlation and lagged analyses remain exploratory, but they show that these pair-level properties can be measured and compared across conditions. It may be possible to recover some similar information by splitting a single recorded sequence after the fact, but whether that gives the same kind of information has not been established. It is something we are exploring.

The validation analyses tested whether the architecture and its diagnostic distinctions behaved as intended under known conditions. In the preregistered artifact-injection grid, 30 of 32 statistic-specific cells met the prespecified cancellation criterion, two were indeterminate because of limited real Baseline precision, and none failed. A separate held-out synthetic validation showed that frozen diagnostic rules could distinguish constructed marginal and relational changes, localize injected temporal relationships, and identify a programmed change in local relational stability. Together, these tests show that the implemented pipeline can recover the statistical structures it was designed to detect under the controlled conditions examined. They do not establish universal protection against naturally occurring artifacts or identify the cause of an empirical difference.

Study 1 was designed primarily as a pilot of the architecture and processing pipeline. Its short sessions, highly unequal participant contributions, retrospectively selected subgroup, and incomplete separation of provider and collection-period effects limit population-level inference. The paired hit-rate analyses also lacked the resolution required to detect effects on the scale reported in much of the previous RNG literature. Even with those limitations, the pilot showed that the architecture could be implemented, its diagnostic components could be examined separately, and weaknesses in the original design could be identified clearly enough to address prospectively.

The follow-up study will test the candidate ordering pattern under stronger conditions. Longer fixed sessions, repeated participation, contemporaneous Baseline collection, dependence-aware inference, and prespecified analyses address several of the principal limitations of Study 1. The primary confirmatory question will be the direct, two-tailed Human-versus-Baseline Subject-stream H_{RS} comparison. Paired-differential and PCS analyses will provide the mandatory diagnostic decomposition needed to understand how any direct difference is expressed across the two streams. Relational analyses will remain exploratory and will examine whether the candidate variability and temporal-relationship patterns warrant further prospective development.

More broadly, PDP reflects a simple methodological principle. When expected effects are small and alternative explanations are difficult to separate, greater sample size improves precision but does not automatically provide more information about the structure of an observed difference. A matched companion measurement from the same acquisition adds that information. PDP keeps the conventional direct comparison, adds a conditional within-call reference, and makes joint-stream behavior observable. Its value comes from increasing the amount of interpretable information obtained from each acquisition.

For research on subtle, heterogeneous, or contested effects, that additional information changes what can be learned from a result. A null result can be characterized with greater resolution. A positive result can be decomposed to show how Subject and PCS contributed to it. Relational structure can be examined directly and carried forward as an explicit question for future testing. Combined with preregistration, adequate sample sizes, repeated participation, contemporaneous controls, and independent replication, PDP provides a framework for more discriminating investigation of effects that are otherwise difficult to separate from ordinary system variation.

Acknowledgements and Disclosures

In memory of my father, Alfred C. Rester Jr. (1940–1996), nuclear physicist and arms-control scientist, whose work ranged from the Gamma Ray Advanced Detector to the observation of Supernova 1987A, and whose love of discovery inspired my own.

AI-Assisted Research

Large language models were used during the development of experimental code, data processing pipelines, and manuscript drafting. The study design, architectural decisions, and analytic strategy were directed by the author. All generated code and analyses were reviewed and validated against the frozen dataset prior to inclusion in the manuscript.

Data Transparency

All data collection procedures and metrics were defined in the experiment source code prior to analysis. The dataset was frozen using SHA-256 hashing before statistical evaluation. Raw data and full computational notebooks are available in the public repository referenced in the Data Availability section.

Ethics and Informed Consent

The study was conducted by an independent researcher without institutional affiliation, and prospective review by an institutional review board or research ethics committee was not obtained. The study was conducted with reference to the principles of respect for persons, beneficence, and justice described in the Belmont Report (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979). Participation was voluntary and occurred through an online experimental interface. Participants were informed about the study before choosing to participate. The noninvasive computer-based task typically required approximately 3–5 minutes, and participants could leave the study at any time. No names, contact information, IP addresses, or other direct personal identifiers were collected through the experimental interface. Participants were provided with a study-generated session identifier that could be used to request their results or removal of the corresponding session data. Given the brief, noninvasive procedure, voluntary participation, and absence of direct identifying information from the experimental dataset, the study was considered to present limited foreseeable risk. This assessment was made by the author and was not a formal determination by an independent ethics committee.

Conflict of Interest

The author declares no financial interest in any QRNG hardware, software platform, or related commercial product. This work was conducted independently and without external funding.

Data Availability Statement

The raw datasets, the frozen data manifest (SHA-256), and the Python analysis notebooks used to generate the findings in this study are available in the public repository at 10.5281/zenodo.20763628. The experimental source code (JavaScript), including the pre-specified hurstApprox and delta logic, is also provided on GitHub to ensure full transparency and replicability. The artifact-injection validation grid was preregistered at OSF (DOI: 10.17605/OSF.IO/BG9YS).

Supplement

This Supplement retains detail cut from the main text during compression of Sections 3.6 and 3.7. Nothing here changes the status of any result in the main text. Items retained here are exploratory, non-confirmatory, and in most cases already summarized with their conclusion stated in the main text. This Supplement does not introduce new claims.

S1. Full Position, Weighting, Recurrence, and Session-Order Detail (expands Section 3.7.1)

This section provides additional sensitivity analyses of the Human 5+ H_{RS} pattern summarized in Section 3.7.1. The analyses examine within-session position, weighting, recurrence across sessions, influence of individual sessions and contributors, and first-versus-later session behavior.

S1.1 Within-Session Position

Within-session position was first examined by estimating the change in the paired Subject-minus-PCS single-scale H_{RS} score from the beginning to the end of each complete session. The estimated slopes were +0.00553 for Baseline (95% descriptive interval $[-\ 0.00244, +\ 0.01350]$), −0.00178 for AI ($[-\ 0.01235, +\ 0.00929]$), −0.00137 for all Human ($[-\ 0.00843, +\ 0.00533]$), and +0.00008 for Human 5+ ($[-\ 0.01212, +\ 0.01250]$). All intervals included zero and no group showed a resolved within-session trend.

A second, post hoc diagnostic divided each retained session into early blocks 0 through 14 and late blocks 15 through 29. Baseline's early paired-score mean was −0.004065, compared with +0.000583 in its late blocks. The early-minus-late Baseline contrast was −0.004647, with a session-clustered 95% interval excluding zero at the unadjusted level. The corresponding early-minus-late contrasts were unresolved in Human and AI. None of the three condition-specific comparisons remained resolved under the Bonferroni threshold of $0.05/3 = .01670$.

Within Human 5+, the early paired-score mean was +0.005857 and the late mean was +0.002842. The Human 5+ minus Baseline contrast was +0.009921, with a 95% interval of $[+\ 0.002908, +\ 0.021763]$, in the early half and +0.002259 $[-\ 0.005471, +\ 0.014116]$ in the matched late half. The late-minus-early change in the condition contrast was −0.007662 $[-\ 0.019489, +\ 0.004125]$, so the direct comparison did not resolve a difference between halves.

As an additional sensitivity check, comparing all Human 5+ blocks with only the late Baseline blocks reduced the Human 5+ minus Baseline contrast to +0.00377, with a 95% interval of $[-\ 0.00184, +\ 0.01366]$. Together, these diagnostics show that the negative early-session Baseline realization contributed to the magnitude of the observed contrast, without establishing a systematic within-session position effect.

S1.2 Weighting Sensitivity

The Human 5+ minus Baseline paired H_{RS} contrast was evaluated under several weighting schemes. Block weighting produced a contrast of +0.00609. Equal-session weighting produced the same value because every retained session contributed 30 blocks. Giving each of the three Human 5+ contributors equal weight, and then weighting their sessions equally, produced a contrast of +0.00818. The direction of the contrast was therefore unchanged under the weighting schemes examined.

A hierarchy-preserving bootstrap used 10,000 draws. Human resampling selected contributors and then complete sessions within contributors, while Baseline resampling selected complete sessions. Blocks within sessions were retained together. For the block-weighted contrast, the descriptive 95% interval was $[+\ 0.00094, +\ 0.01585]$.

S1.3 Session Recurrence and PI-Influence Sensitivity

The Human 5+ subgroup contained three contributors and 28 complete sessions. Fifteen sessions had positive paired H_{RS} scores and 13 had negative scores. The median session-level paired score was +0.003492, with an interquartile range of −0.007743 to +0.015702.

Leaving out each complete session in turn produced pooled paired scores ranging from +0.003270 to +0.005046, compared with +0.004349 in the full subgroup. The observed direction therefore did not depend on any single session.

The principal investigator contributed 18 of the 28 sessions. Excluding the principal investigator increased, rather than decreased, the block-weighted paired score $(+\ 0.004349\ to\ \ +\ 0.008682)$ and the Human 5+ minus Baseline contrast from $+\ 0.006090\ to\ \ +\ 0.010423$. The positive subgroup contrast therefore did not depend on inclusion of the principal investigator's sessions.

S1.4 First Versus Later Sessions

A descriptive within-contributor analysis compared each eventual Human 5+ contributor's first complete session with their mean across later sessions. The equal-contributor mean later-minus-first change in paired H_{RS} score was +0.004560 but this average obscured divergent individual patterns. Two contributors increased (range −0.013107 to +0.014535) and one decreased. With only three contributors, these data do not resolve a consistent first-versus-later session pattern.

S2. Full Local Relational-Variability and Lagged-Alignment Detail

This section examines whether the variability of local Subject–PCS correlations differs across conditions, how that pattern behaves across window sizes, whether it recurs at the session level or in a separate dataset, and whether lagged Subject–PCS relationships show unusual temporal structure.

S2.1 Local Subject–PCS Correlation Variability

Exploratory analyses examined whether the variability of local Subject–PCS correlations differed between Human and Baseline conditions. Correlations were calculated within nonoverlapping windows of blocks within each session, and the variability of those local correlations was then compared across conditions. The quantity of interest was therefore the stability of Subject–PCS coupling within sessions rather than the mean correlation itself.

The original pilot contained only 30 blocks per session, which limited reliable evaluation to short window sizes. In the full Human population, comparisons with Baseline at $W = 5$ and $W = 10$ were unresolved. A separately collected dataset with 80-block sessions allowed the analysis to extend to $W = 15$ and $W = 20$. In that dataset, lower Human variability was observed at the longer window sizes in the full Human population.

Subsequent exploratory work treated variance narrowing as a pattern across the window-size curve rather than as a finding tied to any single window. This motivates a candidate second hypothesis for Study 2, whose confirmatory or exploratory status remains under evaluation, asking whether lower Human variability persists across several prespecified window sizes, rather than appearing at a single point. Session-level reproducibility will be examined separately as an exploratory question. This formulation was informed by the exploratory pilot and prescreen data and will be evaluated prospectively on new data.

S2.2 Lagged Subject–PCS Coupling

Session-level cross-correlations between Subject and PCS H_{RS} scores were examined at lags from −3 through +3. A positive lag indicates that Subject at block t was correlated with PCS at block $t + k$. The analysis compared Human and Baseline sessions using seven mean comparisons and seven variance comparisons, forming a 14-comparison family. Only the variance comparison at lag+2 survived the Bonferroni threshold of $0.05/14 = .00357$. At this lag, Human correlations had approximately 45% lower observed variance than Baseline ($variance\ ratio = 0.5505$, Levene $p = 0.00110$; hierarchy-preserving sensitivity interval: 0.383–0.808). A sensitivity analysis resampling Human contributors and their complete sessions hierarchically, and Baseline sessions as complete units, found the lag+2 result remained unusual both individually $(p = 0.00170)$ and after maximum-statistic adjustment across the seven variance comparisons $(p = 0.00780)$. A further conservative adjustment covering all 14 mean and variance comparisons gave $p = 0.01560$, so the lag+2 variance result held up under the strictest correction applied. The corresponding mean-correlation difference was unresolved $(+ 0.0354,\ p = 0.154)$, and no other comparison survived family adjustment.

This correction covers only the 14-comparison lag family and does not encompass the broader collection of exploratory analyses conducted in the pilot, so this result should not be read as confirmatory. No mechanism specifically predicted a two-block displacement, and differences in participant composition, collection cadence, score distributions, or temporal structure could affect the variability of session-level correlations.

S3. Ordering Sensitivity and Run-Structure Checks

This section provides additional detail on two exploratory checks used to characterize the single-scale H_{RS} score. Section S3.1 examines how the Human 5+ versus Baseline separation changes when within-block ordering is progressively disrupted while bit composition is preserved. Section S3.2 examines the relationship between H_{RS} and simpler run-based properties of the observed sequences.

S3.1 Shuffle Sensitivity to Within-Block Ordering

The H_{RS} score is calculated from the cumulative-deviation path of a sequence. Reordering the same bits can therefore change the score even when bit composition is unchanged.

For each paired block, the same positional permutation was applied to the Subject and PCS streams within non-overlapping windows. This preserved the bit composition, pairing, and shared positional transformation while changing within-stream ordering. Window sizes were 1, 3, 5, 10, 25, 50, and 150 bits. At $W = 1$, the ordering was unchanged. At $W = 150$, each complete 150-bit stream was randomly reordered. The procedure was repeated 200 times for each nontrivial window size.

The unshuffled KS distance between the Human 5+ and Baseline paired H_{RS}-score distributions was 0.0541. For each nontrivial window size, the table reports the mean KS distance across 200 shuffle realizations.

Table S1. Mean Human 5+ versus Baseline KS distances after composition-preserving paired-stream shuffling

Window size	Mean KS distance
1, unshuffled	0.0541
3	0.0491
5	0.0600
10	0.0628
25	0.0586
50	0.0395
150, full-block shuffle	0.0333

The response was not monotonic across window sizes. The smallest mean distance occurred after full-block shuffling. These results show that the measured separation changed when within-block ordering was altered while bit composition was held constant.

This analysis characterizes the ordering sensitivity of the score. The Human 5+ comparison itself is evaluated in Section 3.6.4.

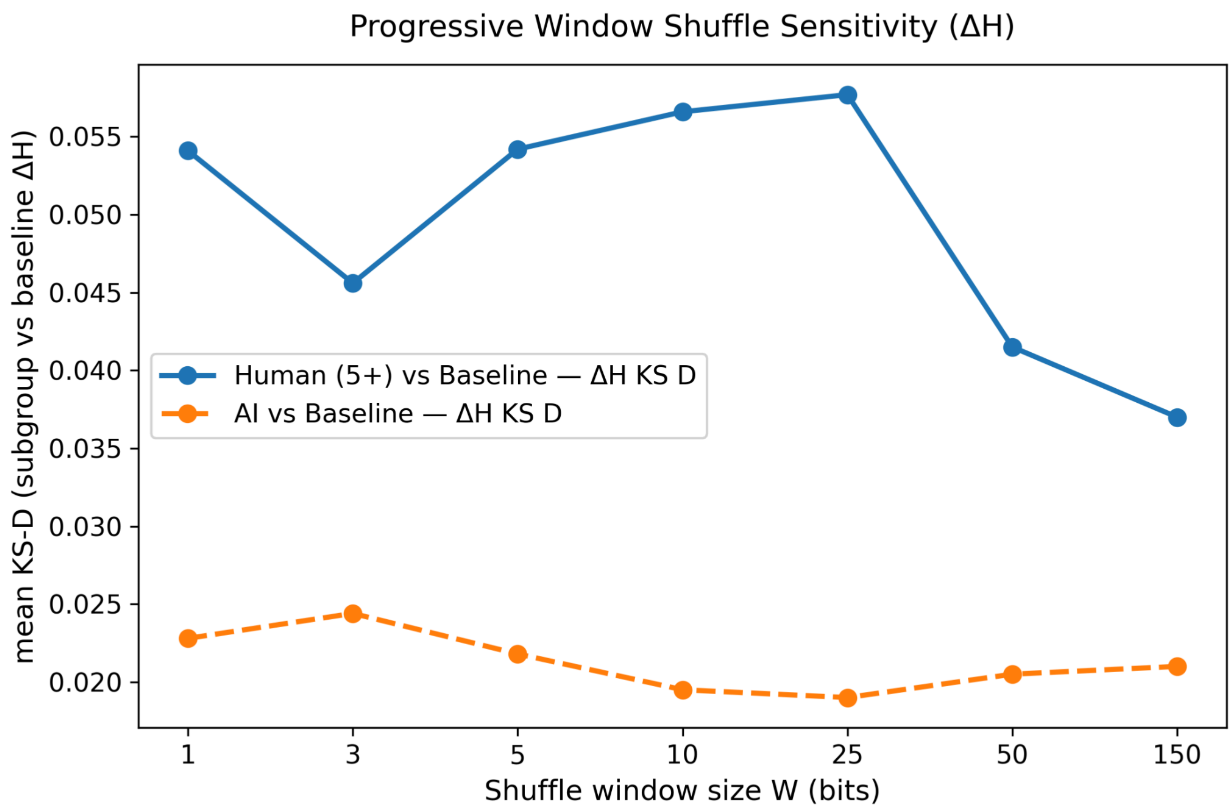

Figure S1. Sensitivity of the Human 5+ versus Baseline distributional separation to within-block shuffling.

The figure shows the mean KS distance between the paired H_{RS} score distributions after the same within-window positional permutation was applied to both streams of each block. Each nontrivial window size was evaluated using 200 realizations. $W = 1$ is the unshuffled reference and $W = 150$ is a full-block shuffle. The figure is descriptive and is not a participant-level hypothesis test.

S3.2 Run-Structure Detail

Run structure was examined to characterize the empirical behavior of the single-scale H_{RS} score in the observed 150-bit sequences. Using the 1st and 99th percentiles of the Baseline Subject-stream H_{RS} distribution as exploratory in-sample cutoffs, Human blocks were classified into lower-tail, middle, and upper-tail regions.

Lower-tail blocks had more runs and shorter maximum runs than typical blocks, whereas upper-tail blocks had fewer runs and longer maximum runs. Mean run counts were 79.44, 75.49, and 68.51 in the lower-tail, middle, and upper-tail regions, respectively. Corresponding mean maximum run lengths were 5.67, 7.58, and 9.60 bits. This pattern is expected because longer same-value runs tend to produce larger cumulative excursions, which increase the single-scale H_{RS} score. The relationship is not an algebraic identity, so this analysis characterizes the score's empirical behavior rather than independently validating it.

A separate paired analysis examined whether Subject-minus-PCS run structure differed between Human 5+ and Baseline while preserving session and participant hierarchy. The Human 5+ minus Baseline contrast was -0.058 runs for run count, with a 95% descriptive interval of $[-\ 0.995, +\ 0.778]$. The contrast was +0.048 bits for maximum run length, with an interval of $[-\ 0.145, +\ 0.279]$, and+0.00089 for run-length entropy, with an interval of $[-\ 0.01379, +\ 0.01772]$. All intervals included zero. These simpler run-based measures therefore did not reproduce the Human 5+ paired H_{RS} contrast at the available resolution.

Run count was treated as the primary run measure. Maximum run length and run-length entropy were correlated secondary descriptions. The Baseline cutoffs, run metrics, and Human 5+ comparison were exploratory.

S3.3 Additional Bit-Composition Checks

The original analysis compared Subject-stream entropy directly across conditions and found no clear condition difference. As an additional Paired-differential analysis, entropy was calculated separately for Subject and PCS and expressed as Subject minus PCS. Across all Human records, Subject entropy was slightly lower than PCS entropy at n-gram orders 1 through 4, with a larger same-direction difference in Human 5+. AI and Baseline remained near zero and did not show a consistent direction across entropy orders. All hierarchy-preserving intervals included zero. The consistently negative Human 5+ estimates therefore represent a weak same-direction pattern that is compatible with sampling noise and does not establish a difference in entropy or short-pattern structure. The analysis nonetheless illustrates how the paired architecture can compare same-call stream properties even when the direct Subject-stream comparison does not clearly differ across conditions.

S3.4 Alternative Ordering and Estimator Cross Checks

The Human 5+ versus Baseline contrast was also examined using alternative measures of sequence structure. Lag-1 autocorrelation showed no resolved Human 5+ minus Baseline contrast in either the Direct comparison $(-\ 0.001627,\ 95\%,\ CI[-\ 0.005044, +\ 0.001501])$ or the Paired-differential comparison $(+\ 0.000112, 95\%, CI[-\ 0.004811, +\ 0.007279])$. N-gram entropy at orders 1 through 4 likewise showed no resolved contrast in either analysis family, and the entropy-rate slope contrasts were unresolved. These measures did not reproduce the H_{RS} pattern at the available resolution.

DFA was examined separately as a cross-check on whether the observed direction depended specifically on the single-scale H_{RS} estimator. DFA showed the same directional Human 5+ pattern, but with a smaller standardized difference, $d = 0.065$, compared with $d = 0.095$ for H_{RS}, and an unadjusted $p = 0.0874$. The shared direction suggests that the observed pattern was not confined to the H_{RS} calculation, while the smaller magnitude indicates that the size of the contrast depended on the metric used.

S4. Session-Count Thresholds and Stream-Level Decomposition

This section asks whether the Human 5+ result is an artifact of which session-count threshold was chosen, and how the Subject and PCS streams contribute to the observed paired contrast.

The Human 5+ threshold was initially examined because participants recruited through the Reddit remote-viewing community had been encouraged to complete five to ten sessions. The 2+, 3+, and 5+ thresholds were subsequently examined as retrospective comparisons.

These thresholds were nested. Participants in the 5+ group were also included in the 2+ and 3+ groups. The comparisons therefore did not represent independent participant groups. The disjoint 2–4 group was examined separately as a follow-up comparison.

Table S2. Session-count threshold comparison

Human session count group	Contributors	Sessions	Blocks	Mean paired H_{RS}	Unadjusted KS p
2+ sessions	10	47	1,410	+0.00389	.0577
3+ sessions	6	39	1,170	+0.00341	.0496
5+ sessions	3	28	840	+0.00435	.0398
2–4 sessions, disjoint follow-up	7	19	570	+0.00322	.244

Table S2. The 2+, 3+, and 5+ groups are nested, overlapping thresholds rather than independent groups. Their block-level KS values are retained as unadjusted descriptive results. The disjoint 2–4 group excludes the three 5+

participants and was examined separately as a follow-up comparison. The composition-conditional order-randomization analysis reported in Section 3.6.4.2 provides the documented threshold-search calibration.

The 5+ comparison was the strongest of the nested thresholds, but because the same three 5+ participants contributed to all three nested groups, the pattern across the 2+, 3+, and 5+ thresholds should not be interpreted as an independent dose-response gradient. The disjoint 2–4 group, which excludes those three participants entirely, did not show a clear distributional difference from Baseline ($p = .244$) and the pattern did not extend to Human participants outside the specific subgroup that generated it.

Separate Subject and PCS comparisons were also examined to describe how the paired contrasts arose. The paired t-test and label-swap test in Table S3 evaluate the same mean paired HRS contrast against Baseline. They are alternative analyses of the same underlying contrast rather than independent evidence.

Table S3. Stream-level directional pattern by cohort

Cohort	p_t	p_{swap}	Subject vs Baseline	PCS vs Baseline
Human, 1 session	0.287	0.287	↑ ns	↔ ns
Human, 2–4 sessions	0.090	0.085	↑ marginal	↔ ns

Human, 5+ sessions	0.0147	0.014	↑ marginal	↓ marginal
AI Agent	0.084	0.086	↔ ns	↔ ns

Table S3. The paired t-test and label-swap test evaluate the same mean paired HRS contrast against Baseline. The label-swap test randomly exchanges Subject and PCS labels within blocks. The Subject and PCS columns show the direction of each stream relative to Baseline based on separate stream-level tests: ↑ indicates higher, ↓ indicates lower, and ↔ indicates no consistent direction. "Marginal" indicates $0.05 < p < 0.10$, $ns = p \geq 0.10$.

The tests in Table S3 treat blocks as independent and are reported as exploratory descriptions with unadjusted p-values. The opposed Subject and PCS directions in Human 5+ describe the arithmetic components of the paired contrast, not separate confirming evidence or a stream-specific mechanism.

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